

Event-Based Decentralized Approach of Root Zone Soil Moisture Increase Dynamics

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Abstract

Root zone soil moisture (RZSM) is critical for irrigation management, as it directly affects plant water availability, crop growth, and irrigation scheduling. However, modeling RZSM is challenging due to the high variability and nonlinearity of soil moisture patterns. Physically based models, such as those solving Richards' equation, offer detailed soil dynamics but require extensive hydrological parameters and significant computational resources. In contrast, statistical approaches are more efficient but lack physical interpretability. This study proposes an event-based framework that models soil moisture increases following individual water input events (e.g., precipitation, irrigation). Water balance models are designed to simulate moisture changes within each soil layer, while an XGBoost ensemble captures interlayer interactions. By embedding the machine learning model within physically structured equations, the approach ensures both accuracy and interpretability. The model was applied using soil moisture sensor data collected from weather stations with and without crops in Florida. Results show strong accuracy in event-pattern identification and the modelling of event-scale duration and magnitude behavior across locations.

Keywords

event-based modeling, layered water balance, root zone soil moisture, XGBoost

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Introduction

Ensuring sustainable agricultural production is a critical global challenge, intensified by environmental pressures and the severe limitation of natural resources, especially water. Efficient water management has become paramount, fundamentally transforming traditional practices. For instance, in regions of the southeastern United States, particularly Florida, urbanization and diverse water demands have led to increasing agricultural water scarcity, necessitating advanced water use prediction models (Qi et al., 2020). Current irrigation often wastes water due to a lack of information on subsurface moisture conditions, which rely on surface ground moisture (Orloff et al., 2003). Soil moisture governs the exchange of water movement and redistribution within the root zone, serving as a key parameter for agricultural processes. Root zone soil moisture (RZSM) directly represents the water availability for plants. Both deficits and excesses significantly impact crop growth and yield. Therefore, accurate RZSM prediction is important for optimizing irrigation scheduling, maximizing water use efficiency, and enabling proactive water resource allocation. These efforts directly support agricultural sustainability.

However, RZSM is complex to predict. This is due to its dramatic spatial variability and nonlinear dynamics, influenced by external water input, internal soil properties, topography, and crop water absorption. These multi-scale impact behaviors make the modeling of RZSM a challenge in agricultural science and practice.

The vertical heterogeneity of soil adds more complexity to RZSM prediction. For example, a low permeability intermediate layer (e.g. coarse sand beneath finer soil) can be a capillary or barrier that blocks moisture retention and redistribution. Recently, a study has shown that when modeling soil moisture in a layered model, vertical water movement is highly dependent on depth, texture, and initial moisture status (Z. Zhang & Huang, 2021). Soil in different depths typically consists of layers, each possessing distinct texture, porosity, and hydraulic conductivity. For instance, a middle layer (around

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4–6 in underground) with low permeability can act as a capillary barrier. This slows down the infiltration rate and results in a different moisture retention level for both the layers above and below it (Y. Wang et al., 2025). These interlayer connections make traditional statistical approaches, which often treat the root zone as one bucket, and fail to accurately model the layered RZSM due to their inherent lack of depth-dependent dynamics. While high-resolution simulations of soil physics by physics-based models (e.g. solving Richards' equation) can capture core short soil interlayer connectivity, they require detailed discretization and environmental parameters. These requirements are impractical for many agricultural stakeholders, particularly small-scale farmers, who often lack the budget and equipment to collect environmental parameters such as soil hydraulic functions (e.g. water retention and conductivity), physical soil layer structures, and detailed boundary conditions. Obtaining such data requires laboratory analysis and specialized field equipment, which small-scale farmers often cannot afford or access.

This reveals a fundamental trade-off. Physically based models (e.g. Richards' or Hydrus) ensure physical realism but are computationally intensive and sensitive to uncertain inputs. In contrast, data-driven models (e.g. statistical methods, machine learning algorithms) do not require detailed environment parameters and can capture complex patterns but lack physical grounding and interpretability due to those models learning from data without a physical rule guide. This leaves a gap in current RZSM modeling for practical agricultural applications. Therefore, a hybrid approach capable of preserving physical dependence while effectively learning complex data correlations is needed.

Previous studies have applied data-driven models to multi-layer soil moisture for irrigation scheduling. For example, a study (Adeyemi et al., 2018) used dynamic neural networks to predict multi-depths soil moisture dynamics for control of irrigation, and another study (Gu et al., 2021) developed a soil moisture neural network model to support irrigation scheduling decisions. These approaches demonstrate the performance on multi-layer soil moisture modeling, but without explicitly define physical rules or water balance between layers. In contrast, our framework adopted XGBoost models within layered water balance structure, so that interlayer water flow and soil moisture interlayer changes remain mass-conserving while still allowing flexible, nonlinear responses. An event-based modeling strategy is proposed to embed physical constraints directly into a machine learning framework. Within this structure, a series of XGBoost models is aligned with multilayer water balance equations, where each model handles a distinct layer while remaining intrinsically linked through the water balance equation. This hybrid architecture enables interpretable, physically consistent, and computationally efficient RZSM prediction.

Data and Methodology

Scope of Study

This study is part of a broader project (2021–2026) to develop high-resolution RZSM maps that inform precise

irrigation decisions and water conservation practices in vegetable farms in Florida, US. The crop of interest is sweet corn in a breed (Sweet Corn Hybrid: se/sh2) that roots down to approximately 11 in based on on-site observation. Data collected in this project is in hourly time series for weather data and 15 min timestep for soil moisture data gaged for the entire root zone, obtained from 31 observation sites in the Florida Automated Weather Network (FAWN; naturally impacted condition) and three sweet corn farms in Florida (farming operation impacted condition). Figure 1 shows the geographical distribution of the FAWN and farm sites along with a brief illustration of the precipitation events (see section 2.4.5) that will be used for modeling in this study.

The focused area is characterized by its sub-tropical climate and sandy soil, influencing the design of the methodologies. The local weather is characterized by short-term, intensive late afternoon convections and strong solar radiation. The soil type is associated with high infiltration and percolation capacity. Both characters indicate prompt soil moisture responses to surface water input, leading to relatively prompt interactions across soil layers and high alignments of change phases across root zone soil compared to other loamy or clay soils. Thus, the RZSM dynamics are divided into two phases, increase and decrease, for individual modeling. This study focused on modeling increase phases of soil moisture using machine learning methods directed by physically based water balance models. The decrease phase will be modeled using data-driven pattern fitting methods to determine the profile of the specific drying process in a companion paper (Zhang & Yu, in progress).

Data

This study utilizes two distinct datasets. Dataset 1 comprises weather and soil moisture data collected from 31 locations within the FAWN. This dataset originates from weather stations without planted crops. Dataset 2 includes weather and soil moisture data from three farms planted with sweet corn situated within Florida. The weather data for those farms is acquired from the nearest weather station of FAWN. Soil moisture was measured in soil water amount (equivalent water depth (mm)) at 2, 6, and 10 in below the ground using soil moisture probes (Sentek Drill & Drop™, Sentek Technologies, Adelaide, Australia; See Appendix for details). To facilitate the water-balance modeling, the key measurements for water balance model, precipitation and soil moisture values, are converted to an equivalent water depth (mm). The dataset structure is shown in Table 1.

All soil moisture and precipitation data were subjected to a quality control procedure before event extraction. The quality control procedure removed low quality data that is caused by sensor malfunctions or show unrealistic values (e.g. negative volumetric water content, persistently saturated readings under clear weather) with multiple data quality check procedures (e.g. outlier detection, pattern misalignment, correlation across different measurements, etc.).

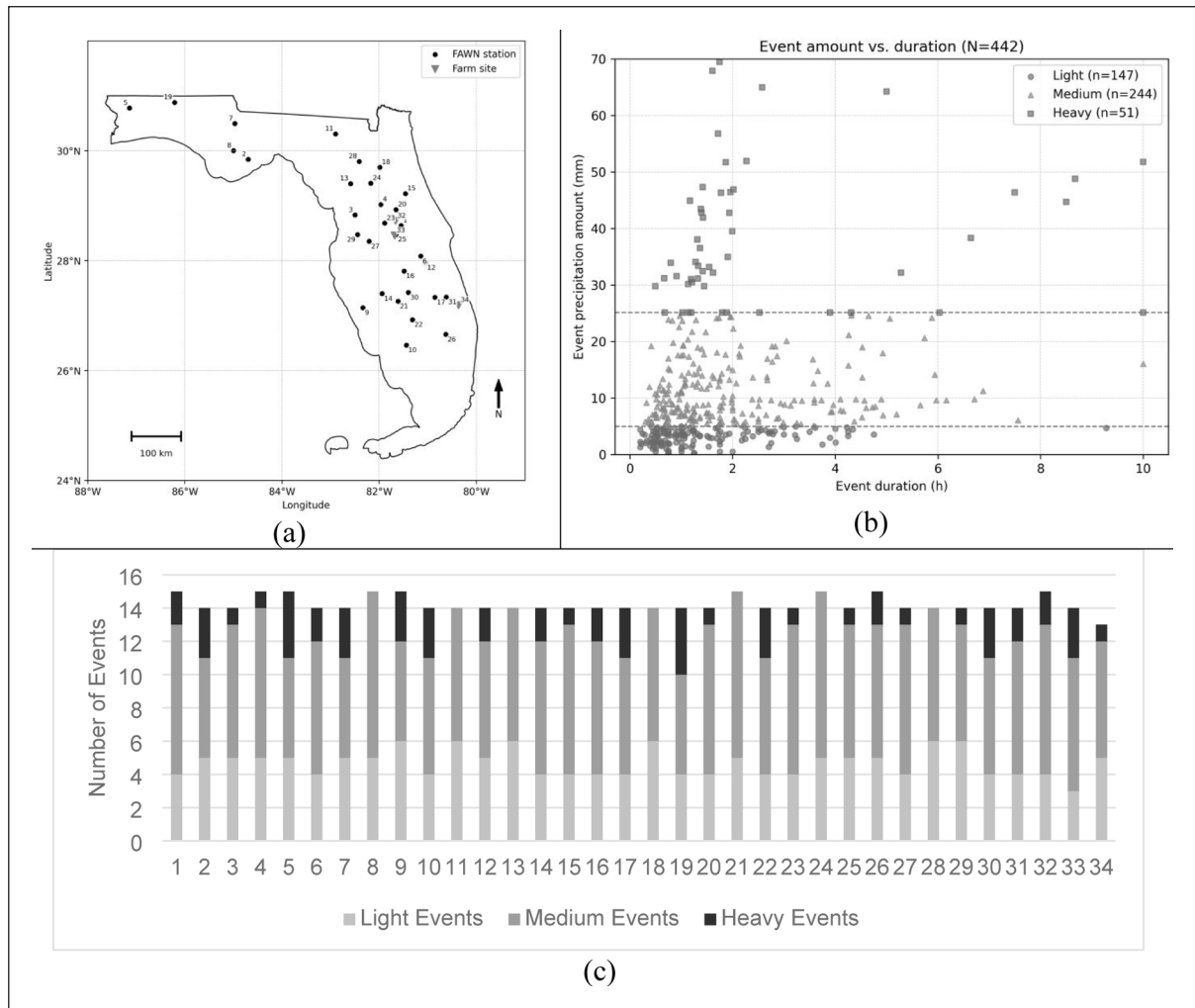


Figure 1. (a) Study area in Florida, USA, showing the geographic distribution of selected FAWN stations (black circles) and farm sites (gray triangles) with in-situ soil moisture sensors used in this study. The numbers of precipitation events are labeled to the corresponding sites, (b) scatter plot of event precipitation amount (mm) versus event duration (h) for all events ($N=442$). Points are grouped into light, medium, and heavy classes using the dashed threshold lines, and (c) the distribution of different precipitation events (see definition in section 2.4.5) for each sites.

Table 1. Dataset Used for Soil Moisture Dynamic Modeling.

Dataset	Source	Locations	Crop presence	Period	Measured variables & units
1 (FAWN)	Florida automatic weather network	31 weather stations across Florida	None	2023–2024	Precipitation & RZSM (mm)
2 (Farm sites)	In-situ sensors	3 farms in Florida	Sweet Corn	2023–2024	RZSM (mm)

Determining Soil Moisture Increase Event

Soil moisture increases are driven by external water input events (e.g. precipitation or irrigation). Modeling soil moisture increases dynamics on an event basis makes it possible to isolate and analyze the fast infiltration and water redistribution, avoiding interference from soil moisture drying (e.g. evaporation). We define soil moisture increase events based on precipitation events determined

by a 3-hr interevent dry period (see Appendix for details). The start of a precipitation event triggers a soil moisture event. The soil moisture event concludes when moisture levels peak, not necessarily when the precipitation ends. Figure 2 shows the workflow for identifying soil moisture increase events corresponding to water input events. Each detected event is characterized by three components: the start timestamp, end timestamp, and the accumulated water input amount during the event. The start timestamp

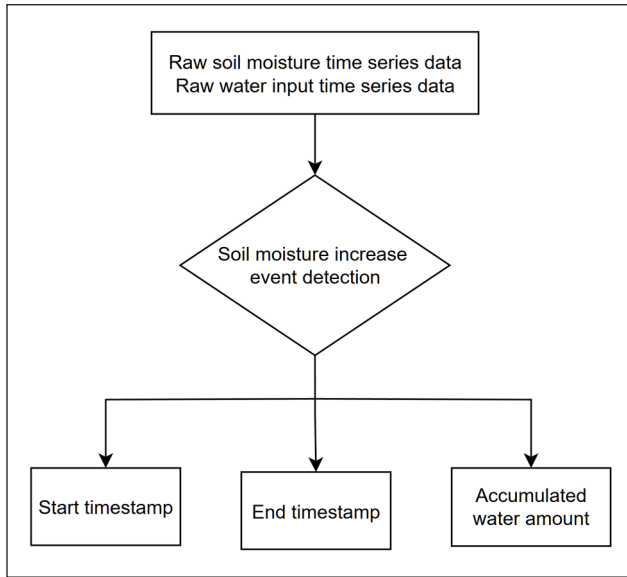


Figure 2. Soil moisture increases the event preparation workflow.

Note. The process begins with raw soil moisture and water input time series data. Events are identified based on defined criteria, and key outputs include the event's start and end timestamps and the accumulated water input.

is defined as the first upward deviation in soil moisture after the water input begins. The end timestamp is set when soil moisture reaches its peak or begins to decrease. This structured data format prepares data for subsequent model establishment.

For modeling purposes, the event characteristics will be used to extract key features, including initial soil moisture (corresponding to start timestamp), water input amount and duration (associated with water input events), soil moisture increased amount (based on end timestamp), and soil moisture increased duration (difference between end and start timestamps), for building a RZSM model. Before training, the combined event data are randomly split into training and validation subsets using an 80% to 20% ratio.

Modeling Framework

This section presents a modeling approach that integrates the physical water balance equation with a machine learning algorithm to estimate soil moisture increase due to surface water input. The model simulates how surface input redistributes vertically across soil layers during short infiltration events. Layered water balance equations provide structural inputs to the XGBoost model, where soil moisture change in each layer is estimated as a function of incoming water, antecedent moisture, and inter-layer connectivity.

According to Florida weather pattern and collected weather data, the precipitation durations are typically range from half to 3 h (Figure 1), and daily reference evapotranspiration are 3.5 to 4.5 mm/day under humidity Florida environment (Irmak et al., 2003; Martinez & Thepadia, 2010). Thus, two key assumptions underline this model

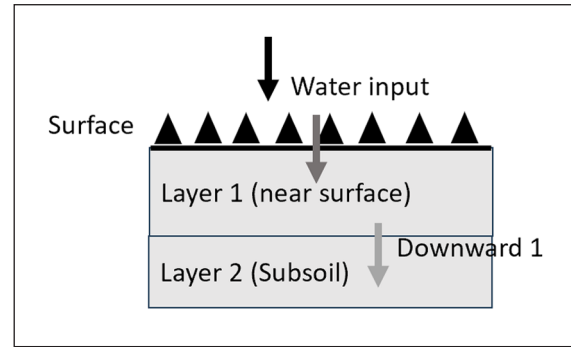


Figure 3. Conceptual diagram of soil moisture infiltration.

Note. Water input at the surface first infiltrates into Layer 1 (near-surface soil), followed by percolation into Layer 2 (subsoil), indicated by the downward arrows. The labeled "Downward 1" represents the moisture transfer from Layer 1 to Layer 2.

framework. First, vertical flow dominates during the short increase phase in sandy soils (Porporato et al., 2002; Western et al., 2004), so that horizontal water movements can be neglected. Second, evaporation, crop water uptake, and drainage are negligible during this period since the duration of infiltration is too short for evaporation to significantly affect the water balance (Vereecken et al., 2008; Zreda et al., 2012). Additionally, since this study is for irrigation management, which does not require fine temporal resolution, we assume that the detailed fluctuations of soil moisture during a precipitation event period are generally not critical for irrigation scheduling and can be ignored for modeling. Instead, the study should emphasize simulating the initial and end soil moistures corresponding to a water input process. This approach is consistent with Singh's soil moisture event study (Singh et al., 2021) that quantify rainfall driven responses using change magnitude and response time rather than the complex non-linear soil moisture characteristics.

Water Balance Model. The water balance model in this project is designed by soil layers. Figure 3 shows how water redistributes when the water is input into the soil surface (Layer 1: Topsoil, 2 in). Under the assumption of negligible horizontal water movements, evaporation, crop water uptake and drainage, water is directly input into the soil and redistributed in two ways, either being retained in Layer 1 based on its soil water holding capacity or percolating downward into the subsoil layer (Layer 2) as Downward 1.

These two ways of water redistribution are represented by water balance equation (1).

$$\text{Water input} = \text{Downward}_1 + (\theta_{1,\text{end}} - \theta_{1,\text{init}}) \quad (1)$$

From equation (1), the total amount of water is conserved, which indicates that the water input into the soil on the left side of the equation and the right side of the equation represents the water transferred from layer 1 to layer 2 (Downward_1) and the water remaining in layer 1 ($\theta_{1,\text{end}} - \theta_{1,\text{init}}$).

In practice, *Water input*, $\theta_{1,init}$ is known information. Since this soil moisture increase phase modeling approach is part of the soil moisture dynamic modeling approach that covers both soil moisture increases and dry phase, the $\theta_{1,init}$ is provided by the dry phase model (Zhang & Yu, in progress) and used in this increase phase model. Therefore, to calculate the water flow out of layer 1 *Downward₁*, need to know the $\theta_{1,end}$, which is the layer 1's soil moisture content after infiltration diminished.

Now based on the information we have, the variables can be used as input parameters are:

$$x_1 = \{\theta_{1,init}, P_{amt}, P_{dur}\} \quad (2)$$

Where $\theta_{1,init}$ is layer 1 initial soil moisture, P_{amt} is amount of water input, P_{dur} is the duration of water input.

Based on the input parameters x_1 , investigate a model $f(x_1)$ to predict $\hat{y}_1$: the layer 1's soil moisture condition after infiltration diminished.

The model predicts two outputs:

$$\hat{y}_1 = f(x_1), \text{ where } \hat{y}_1 = \begin{bmatrix} \Delta\hat{\theta}_1 \\ \Delta\hat{t}_1 \end{bmatrix} \quad (3)$$

Where $\Delta\hat{\theta}_1$ is the layer 1 soil moisture increased amount, $\Delta\hat{t}_1$ is the layer 1 soil moisture increased duration. The duration of the moisture increases $\Delta\hat{t}_1$ captures the temporal characteristics of infiltration, indicating how long it takes for water to move from the surface into Layer 1.

Then the $\theta_{1,end}$ in equation (1) can be calculated as:

$$\theta_{1,end} = \theta_{1,init} + \Delta\hat{\theta}_1 \quad (4)$$

Submitted equation (4) into equation (1) gets:

$$Downward_1 = Water\ input - \Delta\hat{\theta}_1 \quad (5)$$

Equation (5) shows the water flow intuitively, the downward out water from layer 1 equals to total external water input minus the moisture remaining in layer 1.

The specific water balance models for Layers 2&3 are similar to Layer 1 and provided in detail in the Appendix.

The Interconnection of the Three Water Balance Equations. The interconnection of water balance equations for three soil layers are shown in Figure 4 in terms of parametrization for modeling soil moisture dynamics. The surface layer (2 in) is the layer directly accepting the external water input and antecedent moisture. The model returns the surface layer soil moisture increase (amount and duration) and, by water balance, the downward outflow from Layer 1 to Layer 2. Layer 2 (6 in) is then driven by this downward water and Layer 2 antecedent moisture, the model for Layer 2 then outputs its soil moisture increase and the residual downward water to Layer 3. The deepest layer (10 in) responds to the water downward from Layer 2 and its antecedent moisture, producing its soil moisture increase and the residual percolation beneath the root zone. This

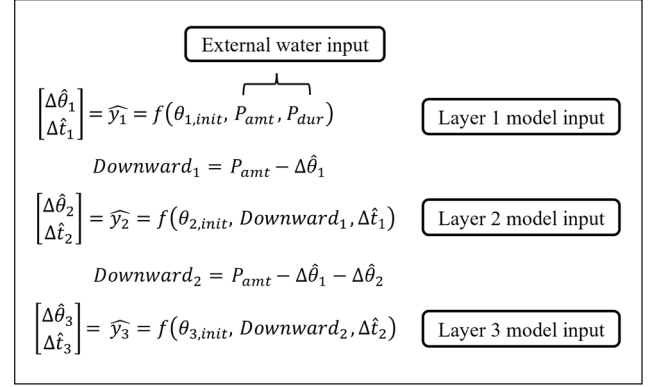


Figure 4. Schematic representation of the hierarchical interconnection of soil moisture responses across three soil layers (2 in, 6 in, 10 in).

top-down structure enforces water mass conservation and encodes the physical dependence on the retention and percolation behavior of overlying layers.

Modeling Approach. This project employs XGBoost to simulate the soil moisture dynamics based on the framework described in 2.4.2. The detailed methodology selection is provided in the Appendix.

XGBoost is a scalable implementation of gradient-boosted decision trees. In gradient boosting, an ensemble of shallow regression trees is built in sequence, where each new tree “learns” to correct the errors (residuals) of the previous ensemble. The predictions are additive: the model starts with an initial estimate (e.g. average value) and sequentially adds trees that predict the remaining error. This technique allows the model to gradually refine its predictions, capturing complex patterns in the data. In addition, XGBoost includes a regularized objective function that penalizes model complexity, which helps to prevent overfitting even as many trees are added. XGBoost objective for regression can be written as:

$$L = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

Where $\ell(y_i, \hat{y}_i)$ is a loss function (such as mean squared error) measuring the difference between observed y_i and prediction $\hat{y}_i$. The $\Omega(f_k)$ is a regularization term for each tree f_k . For each farm location, three XGBoost models (one per layer) are trained, corresponding to the three water balance equations.

Decentralized XGBoost Model Ensemble. Most of hydrological modeling employs centralized modeling that provide a single robust model to simulate systems across different sites by using all data for training neglecting the representativeness of each site. Some researchers pointed out that in hydrology and environmental prediction, cross-site transfer depends strongly on hydrologic similarity and site characteristics, and it can be risky when the key controlling attributes (e.g. soil storage, aridity, or other watershed/soil

Table 2. Hyperparameter Recommendation for XGBoost Models for Future New Locations.

Model	$n_estimators$	Learning_rate	Max_depth	Min_child_weight	Gamma	Subsample	Colsample_bytree
Model 1	250	0.015	3	1	0	1	1
Model 2	150	0.015	3	1	0	0.9	1
Model 3	50	0.015	8	1	0	0.8	1

properties) are missing or not represented in the inputs (Chang & Rubin, 2019; Song et al., 2019). Our event dataset is not evenly distributed across locations and the event allocation is very site specific, so a centralized model can be biased toward data-rich sites and conditions, which is a known challenge in hydrologic ML when training data are imbalanced.

Therefore, rather than lumping all data to train a single centralized model for all locations and soil layers, this project adopts a decentralized approach that develops independent models for each location and each soil layer. Such process may reduce the risk of overfitting that a single location's poor or superior data quality or amount influence the model's performance. This logic is consistent with the guidance that performance in multi-site environmental prediction often improves when model explicitly account for site characteristics and process context (Willard et al., 2025).

In this study, there are 31 locations equipped with soil moisture and weather sensors provided by FAWN and on-farm sites. The distribution of these events shows obvious non-uniformity in terms of scattered location of events in Figure 1b, and the segment of event types across locations in Figure 1c. It is more obvious in heavy events, which are distributed in two separated clusters over the duration in Figure 1b, and show noticeable discrepancies across locations Figure 1c. If using centralized model, the heavy events in the left cluster (<4 hr) will overfit the model making long heavy events not represented. In addition, the locations without observed heavy locations may receive an average guess from other locations' events without knowing the confidence of its representative to the location conditions. Therefore, a decentralized modeling framework is used in this study to provide a conservative modeling for every individual location using the corresponding data. In this way, the uncertainties of varying soil properties, weather, and farming operations across different locations will not be counted as factors of bias attributed to overfitting. Instead, the impact of inaccuracy is restricted within very single location due to limited data size. However, this project is expected to serve for a long period of time during which observations could be continuously collected to enrich the data and update the models, improving the accuracy overtime.

At each location, a separate ensemble of three XGBoost models is developed, one for each soil layer (e.g. 2, 6, and 10 in). This gives a total of 93 XGBoost models (31 locations $\times$ 3 layers). By designing models in this decentralized way, each XGBoost model ensemble focuses on the unique soil and weather dynamics at its specific location. In a

single location, the three-layer-specified XGBoost models are designed according to the connected water balance equations in Section 2.4.2.

But due to the limited data volume and the decentralized design, to mitigate the potential problem of accuracy, each model has been individually tuned using Bayesian optimization to find the localized suitable XGBoost hyperparameters to improve the robustness of the model (see Appendix). However, manually tuning each model becomes a challenge once there are new locations included in the system. To simplify the further implementation of the system, recommendation values for hyperparameters are generated by identifying the most statistically representative value for each hyperparameter based on all existing locations. The recommended hyperparameters are presented in Table 2, and the detailed tuning procedure is provided in the Appendix. The development of the model employs a train-test split approach with 80% of water input events (442 events) from all locations to train location-specific XGBoost model ensembles, and the remaining 20% (99 events) are used for prediction and validation.

Evaluation Strategy. To evaluate the performance of the XGBoost model ensemble, this study uses both qualitative and quantitative metrics. Qualitative metric checks the XGBoost model ensemble's ability to detect soil interlayer connectivity. For example, if a light event showing soil moisture increases in layer 1 may not be qualitatively modeled if it also shows soil moisture increases in layer 3. The quantitative metric focuses on the accuracy of the predicted change in soil moisture amount and duration. Three representative types of precipitation events, determined by the depth of water percolation, have been selected to show the evaluation process. If soil moisture increases are observed in all three layers, this event will be defined as a heavy precipitation event, while medium and light events trigger soil moisture increases in layer 1&2 and layer 1, respectively.

The XGBoost model's overall performance is examined by summarizing both qualitative and quantitative metrics for all precipitation types in all locations. The correct detection of whether the soil moisture in each soil layer increases is defined as the qualitative measure of the performance of this model, as the pre-condition to investigate the magnitude of change. The overall quantitative metrics are represented by three evaluation errors: the differences of duration (delta duration), the increased soil moisture (delta increased soil moisture) between predictions and observations, and their product, delta duration $\times$ delta increased soil moisture.

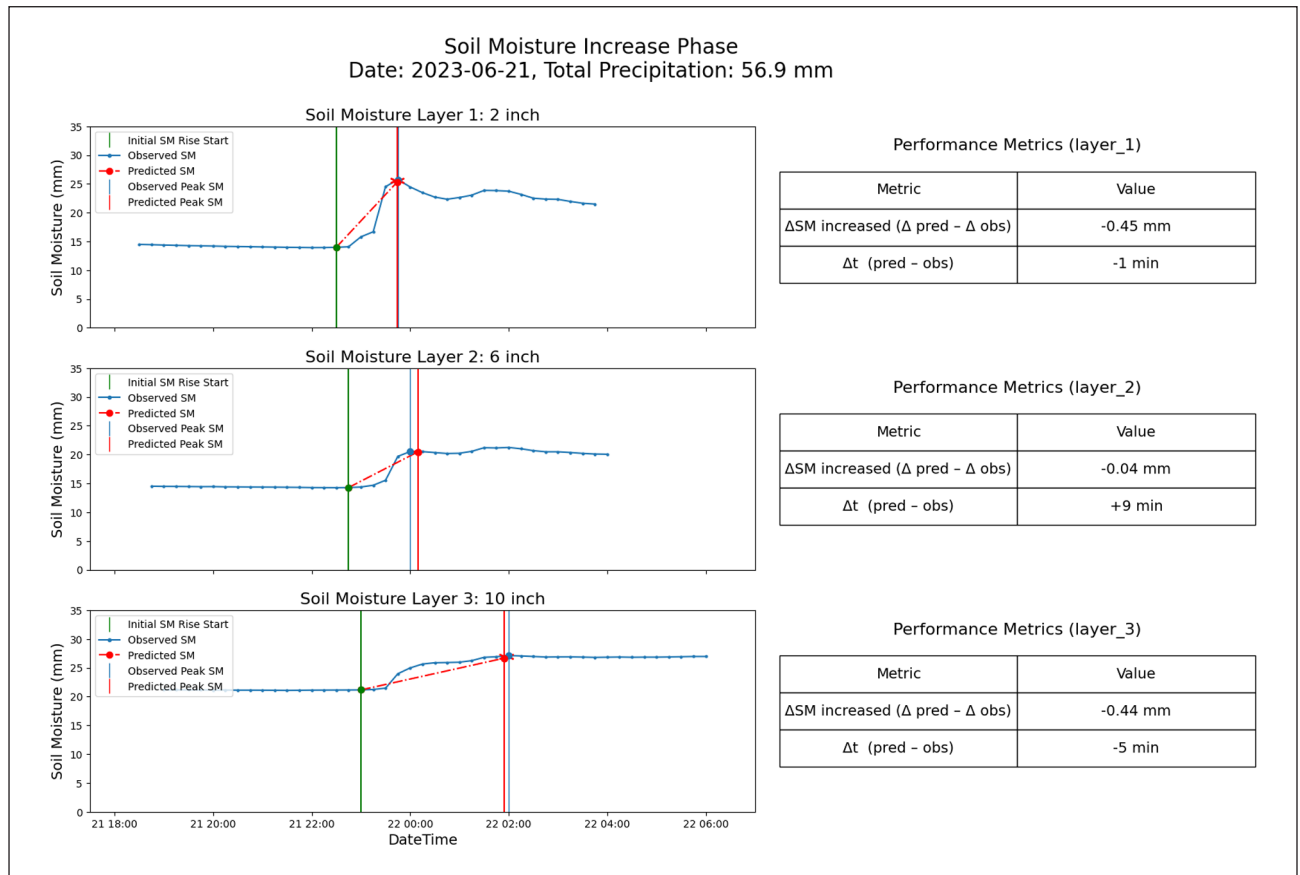


Figure 5. Soil moisture response and prediction performance for a heavy rainfall, 56.9mm, 37.9mm/h.

Note. The left side three subplots correspond to the soil layer at 2-, 6-, and 10-in depths, which display observed soil moisture (blue line with markers), the XGBoost-predicted event-scale rise (red dashed line with markers), the start of soil moisture increase (green dashed vertically), and the peak of SM for observed curve and predicted curve (blue and red vertical lines). Note that the red dashed segment is a linear interpolation drawn between the predicted start and predicted peak to visualize the predicted ΔSM and Δt ; intermediate within-event (nonlinear) wetting dynamics are not explicitly modeled. Right side tables summarized error metrics for three subplots: ΔSM increased (the difference of increased soil moisture between predicted and observed, in mm of water depth) and Δt (the difference of event duration between predicted and observed, in minutes).

Results

Interlayer Soil Moisture Response (Heavy Precipitation Event)

Figure 5 illustrates the soil moisture response across three layers during a heavy precipitation event on June 21, 2023, that randomly selected from FAWN, with a total rainfall of 56.9mm and an intensity of 37.9mm/h, causing moisture increases in all three layers. From a qualitative perspective, the XGBoost model ensemble correctly captures these increases across all layers. Quantitatively, in the surface layer (moisture_1), the predicted soil moisture increase was 0.45 mm lower than observed, with the modeled value reaching peak 1 min earlier. In the subsoil layer (moisture_2), the predicted increase exceeded the observed value by 0.04 mm, with a 9-min delay in reaching the peak. In the bottom layer (moisture_3), the model underestimated the increase by 0.44mm and predicted the peak 5 min earlier. The observed curves confirm that all three layers responded to the precipitation event, and the model reproduced these patterns with close agreement in both timing and magnitude.

Interlayer Soil Moisture Response (Medium Precipitation Event)

Figure 6 presents the results of a medium precipitation event on June 21, 2023, that randomly selected from FAWN, with a total rainfall of 11.9mm and an intensity of 7.2mm/h, showing noticeable increases in the surface (moisture_1) and subsoil (moisture_2) layers. From a qualitative perspective, the modeled results correctly captured the increasing trends in all layers, indicating an appropriate response. Quantitatively, in the surface layer (moisture_1), the predicted soil moisture increases differed from the observed value by +0.02mm, with the peak occurring 1 min earlier. The differences in the subsoil layer (moisture_2) were 0.08mm lower in soil moisture increase and a 21-min delay in reaching peak moisture. While the deep subsoil layer (moisture_3) also shows slightly increased soil moisture, it is not as noticeable as the other two layers. Despite the relatively medium magnitude of precipitation, the model closely reproduced both the timing and magnitude of the noticeable soil moisture response in the first two layers.

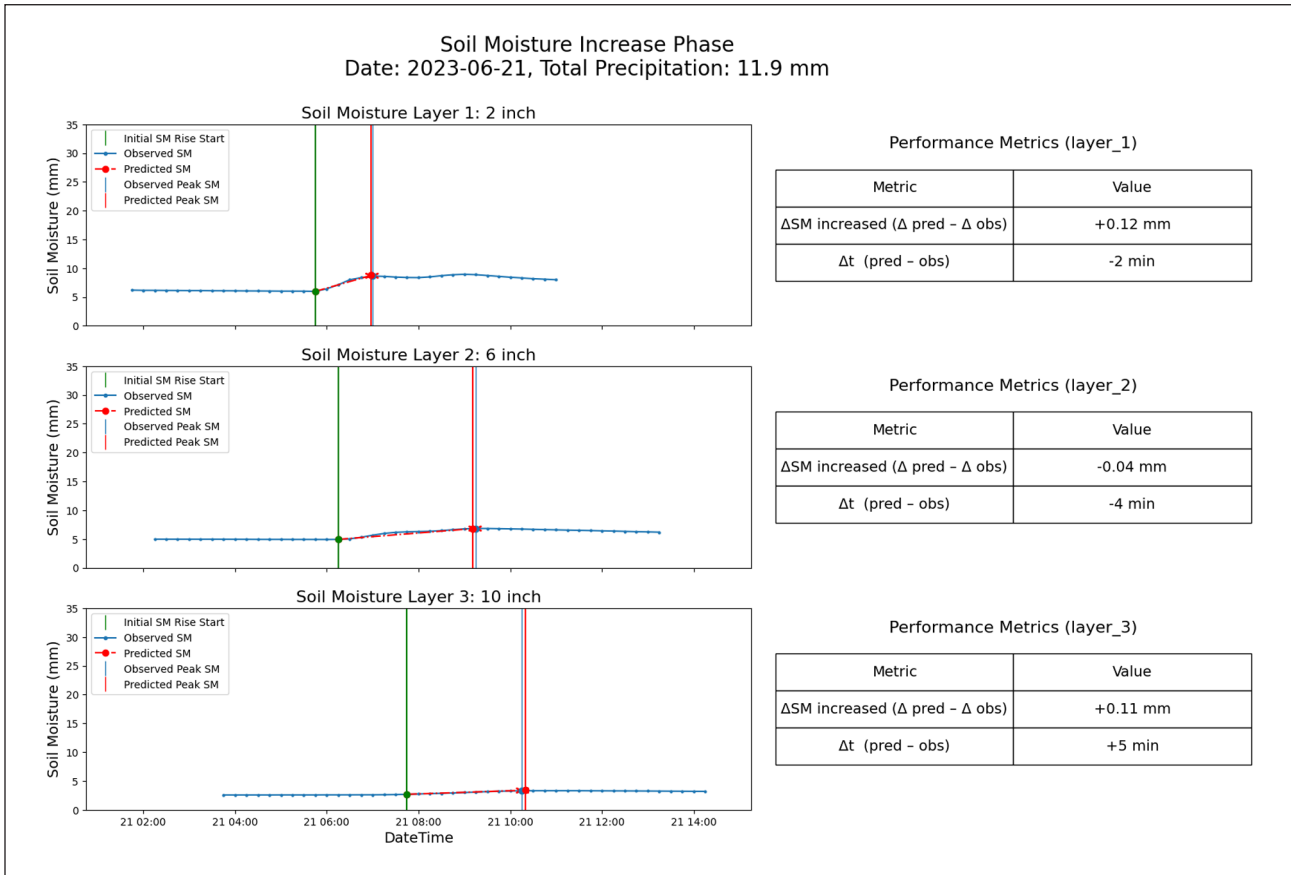


Figure 6. Soil moisture response and prediction performance for medium rainfall, 11.9 mm, intensity 7.2 mm/h. Note. The left side three subplots correspond to the soil layer at 2-, 6-, and 10-in depths, which display observed soil moisture (blue line with markers), the XGBoost-predicted event-scale rise (red dashed line with markers), the start of soil moisture increase (green dashed vertically), and the peak of SM for observed curve and predicted curve (blue and red vertical lines). Note that the red dashed segment is a linear interpolation drawn between the predicted start and predicted peak to visualize the predicted ΔSM and Δt ; intermediate within-event (nonlinear) wetting dynamics are not explicitly modeled. Right side tables summarized error metrics for three subplots: ΔSM increased (the difference of increased soil moisture between predicted and observed, in mm of water depth) and Δt (the difference of event duration between predicted and observed, in minutes).

Interlayer Soil Moisture Response (Light Precipitation Event)

Figure 7 shows the soil moisture response across three layers during a light precipitation event on April 29, 2023, that randomly selected from FAWN, with a total rainfall of 5.7 mm and an intensity of 5.2 mm/h, resulting in a clear increase in the surface layer (moisture_1). Qualitatively, the model ensemble showed a soil moisture increase in the surface layer and negligible changes in the deeper layers. From a quantitative perspective, the differences in soil moisture increase and change duration are 0.06 mm and 2 min in the surface layer (moisture_1). These values are 0.25 mm and 6 min in the subsoil layer (moisture_2), 0.00 mm and 18 min in the bottom layer (moisture_3). Across Figures 5 to 7, the model predictions align well with observations under heavy, medium, and light precipitation events. This consistency reflects expected hydrological behavior, where light rainfall predominantly affects the upper soil layers.

XGBoost Model Performance (All Events)

Table 3 presents the performance of the XGBoost model ensemble from both event-based and layer-based

perspectives. Soil moisture in a layer will be considered as active if its change exceeds the threshold $\Delta SM \geq 0.5$ mm. This threshold is used only for binary pattern identification (active/inactive) and avoids counting small sensor fluctuations as true layer responses.

Based on events (horizontally), interlayer responses in 18 out of 19 heavy precipitation events are accurately modeled, yielding a 95% accuracy rate. In medium precipitation events, 6 out of 42 events were misclassified, resulting in an 86% accuracy. In light precipitation events, the accuracy drops slightly to 87%, with 5 out of 38 events showing unintended moisture increases in layer 2.

Based on soil layers (vertically), the layer 1 model achieved perfect performance, correctly predicting all observed moisture increases (100%). In layer 2, the model identified 52 out of 61 events with increases, corresponding to an 85% accuracy. The layer 3 model had the lowest performance, detecting 15 out of 19 increases, or 79% accuracy. These results indicate a decreasing trend in model accuracy with increasing soil depth, particularly under lower-intensity precipitation.

We interpret Table 3's results with widely cited thematic classification accuracy guidance: an overall identification accuracy larger than 85% is commonly target as satisfactory/

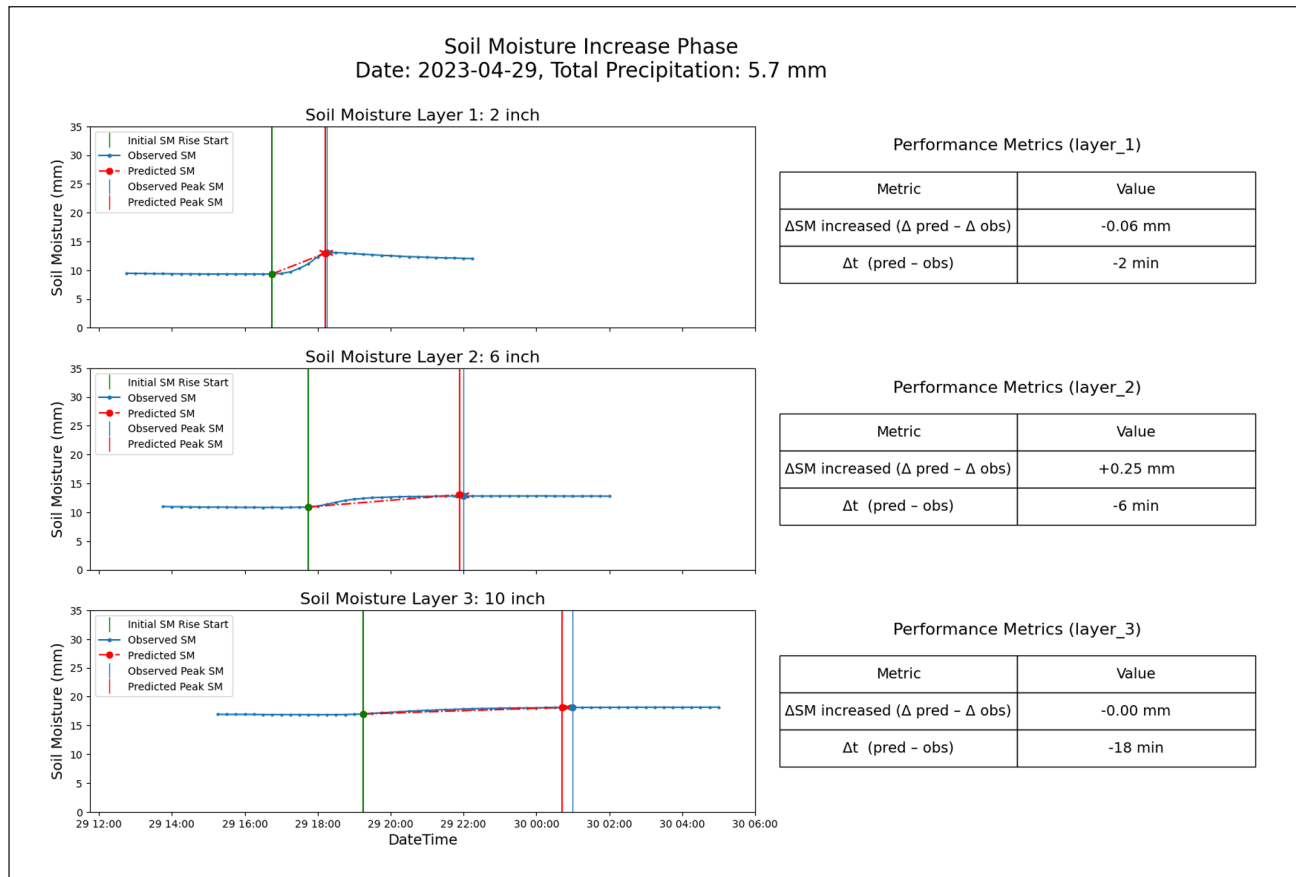


Figure 7. Soil moisture response and prediction performance for light rainfall, 5.7 mm, intensity 5.2 mm/hr.

Note. The left side three subplots correspond to the soil layer at 2-, 6-, and 10-in depths, which display observed soil moisture (blue line with markers), the XGBoost-predicted event-scale rise (red dashed line with markers), the start of soil moisture increase (green dashed line vertically), and the peak of SM for observed curve and predicted curve (blue and red vertical lines). Note that the red dashed segment is a linear interpolation drawn between the predicted start and predicted peak to visualize the predicted ΔSM and Δt; intermediate within-event (nonlinear) wetting dynamics are not explicitly modeled. Right side tables summarized error metrics for three subplots: ΔSM increased (the difference of increased soil moisture between predicted and observed, in mm of water depth) and Δt (the difference of event duration between predicted and observed, in minutes).

Table 3. Comparison of observed and model predicted soil moisture increase responses for heavy, medium, and light precipitation events across three soil layers.

Predicted \ Observed	Layer 1	Layer 2	Layer 3	Event response correct rate
Heavy event	19 / 19	18 / 19	18 / 19	$(1 - \frac{1}{19}) \times 100\% \approx 95\%$
Medium event	42 / 42	39 / 42	3 / 0	$(1 - \frac{6}{42}) \times 100\% \approx 86\%$
Light event	38 / 38	5 / 0	0 / 0	$(1 - \frac{5}{38}) \times 100\% \approx 87\%$
Model error	0	9	4	
Total event	99	61	19	
Layer response correct rate	$(1 - \frac{0}{99}) \times 100\% \approx 100\%$	$(1 - \frac{9}{61}) \times 100\% \approx 85\%$	$(1 - \frac{4}{19}) \times 100\% \approx 79\%$	

Note. Within each diagonal rectangle, the upper number represents the number of events that the XGBoost model predicts exist notable soil moisture increase pattern, the bottom number represents the number of events that observed data show a notable soil moisture increase pattern. The event response correct rate represents the rate at which the XGBoost model correctly identifies that the soil moisture increase pattern exists under heavy, medium, or light events. The fourth row "total event \ model error" represents, in a soil layer perspective, the layered XGBoost model's performance in identifying how many soils moisture increase patterns across all levels of precipitation events. The fifth row, "layer response correct rate," represents the correct rate of the XGBoost model in each layer's performance across all levels of precipitation events.

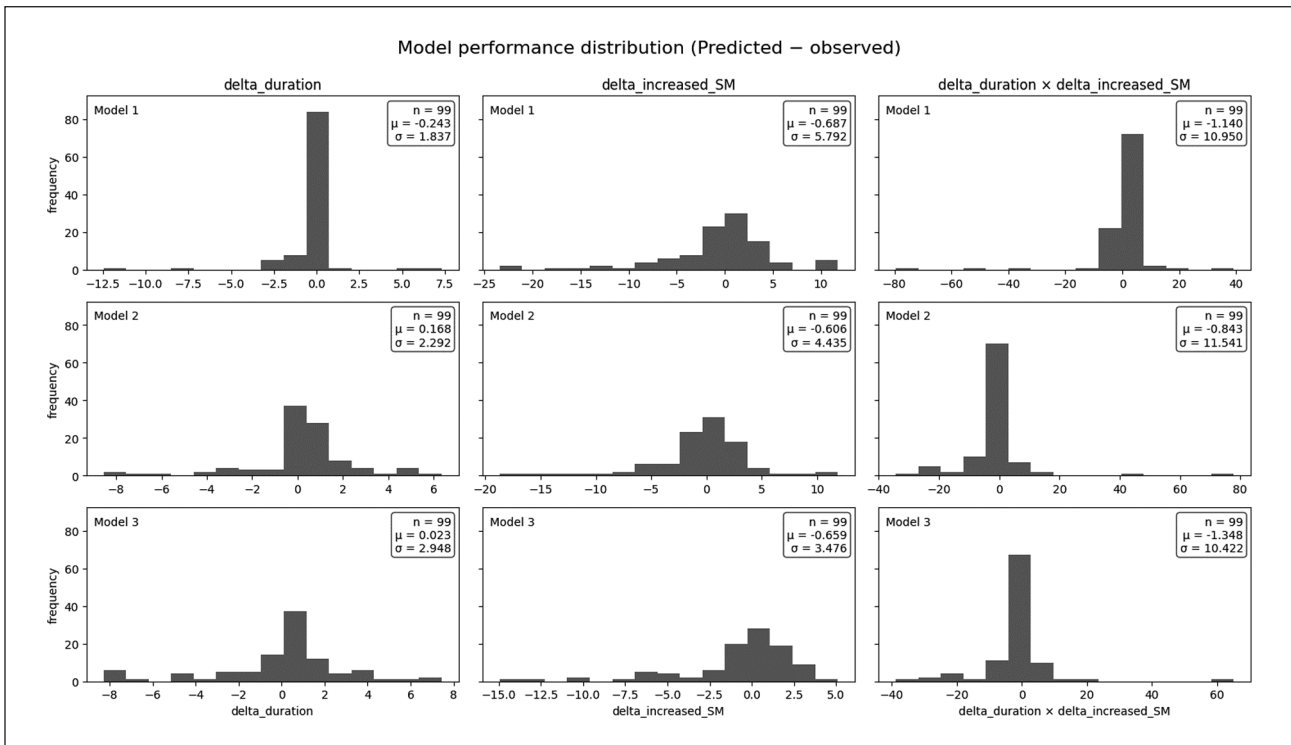


Figure 8. Distribution of XGBoost model ensemble's prediction errors across 99 test events for the three soil layers. Note. Each column corresponds to one soil layer XGBoost model at 2 in (Model 1), 6 in (Model 2), and 10 in (Model 3) soil depths. From top to bottom, each row represents one error metric's distribution across all location events: (1) Δ duration: the timing error between predicted and observed soil moisture increased from initial to peak soil moisture (2) Δ increased SM: the error between predicted and observed increased soil moisture (mm of water depth), and (3) the combined error (Δ duration $\times$ Δ increased SM), this incorporates both timing and amount discrepancies. Each subplot displays the sample size ($n=99$), the mean error (μ), and the standard deviation (σ).

high accuracy, while $\sim 80\%$ is often considered acceptable (Anderson, 1976; Foody, 2008; Olson, 2008). Accordingly, the accuracy of the model performance in identifying increase is high for all summarized conditions except layer 3, which is impacted by the undetected medium events. This is also seen in layer 2 for light events as the increase of soil moisture in such layers are the farthest reach of the corresponding water inputs (layer 2 for light events, layer 3 for medium events) and show a different pattern to the increases under other stronger water input conditions.

Figure 8 illustrates the model's overall performance through histograms of three error metrics: duration error (Δ duration), soil moisture increase error (Δ SM), and their product (Δ duration $\times$ Δ SM). All three error types are clustering around zero, indicating minimal systematic bias. While the mean error (μ) remain small for all models, the standard deviations (σ) are relatively large, suggesting that performance is generally consistent but with some outliers under extreme conditions.

The layer 1 model has the smallest standard deviation in duration error ($\sigma \approx 1.8$), indicating higher accuracy in predicting the timing of surface-layer responses. This may result from simpler model structures and fewer confounding parameters near the surface. However, it also shows the largest standard deviation in soil moisture increase amount ($\sigma \approx 5.8$), reflecting variability in capturing the magnitude of moisture change. This is likely due to greater environmental variability at the surface, while deeper layers are

more stable due to insulation by the above soil. The combined error metric ($\mu \approx -1$, $\sigma \approx 10$) clusters near zero, suggesting that errors in timing and amount often offset each other, resulting in low net bias despite some extreme cases, likely caused by atypical precipitation patterns.

Discussion

Model Performance

For individual events illustrated in results, the XGBoost ensemble effectively captures soil moisture increase dynamics with alignment to known physical patterns. Heavy precipitation events tend to drive water deeper into the soil profile, while light events affect only the surface layer. This response is consistent with hydrological principles: "light rainfall can effectively replenish moisture in shallow soil layers, whereas heavy rainfall can accelerate the soil moisture response" (Ji et al., 2025).

These physical processes explain stratified prediction behaviors. Under heavy rainfall, the steep infiltration gradient and intensive input generate a clear wetting front that moves sequentially through the soil profile (Dai et al., 2022). This makes infiltration and percolation processes more predictable, improving model tractability (Sinha, 2016). As illustrated in Figure 5, these yields pronounce responses that are easier to model. The sharp change mitigates in medium and light events, showing weaker and

shallower responses in the deep layer (Figure 6) and the sub-surface layer (Figure 7). Thus, the model performs best under heavy events, and its learning capacity is increasingly challenged as the event intensity decreases.

This vertical pattern is also evident in the model's layer-wise accuracy. As shown in Table 3, prediction accuracy declines with depth. The surface layer (2 in) shows nearly perfect performance due to immediate hydrologic response to rainfall. The sub-surface layer (6 in) shows lower accuracy (~85%) due to delayed percolation, introducing temporal complexity. This impact was amplified in the deep layer (10 in), exhibiting the lowest accuracy (~79%) and weaker moisture signals. These trends align with soil physics: deeper layers are denser and less permeable, slowing infiltration and enhancing retention. This can also be demonstrated by the increased variance of event durations from surface layers to the deep layer (Figure 8). However, the soil dynamics are also getting more stable in deep layers, which benefits the modeling of the soil moisture increase, with variances decreased from the surface layer to the deep layer (Figure 8). Indicated by the joint error, combining these two dimensions, the modeling performance of the model is relatively similar across all layers. These patterns reflect known physical responses in sandy soils of the region (Y. Zhang et al., 2024).

Quantitative metrics in Figure 8 further confirm this depth-dependent behavior. RMSE results in Figures 5 to 7 show that the surface model (Model 1) achieves the lowest timing error (1.85 min) but has the highest moisture amount error (5.83 mm), likely due to high surface variability. In contrast, the deep layer model (Model 3) shows greater timing error (2.95 min) but lower moisture error (3.54 mm), suggesting slower but more stable infiltration. These contrasting gradients are consistent with the model's design and physical layering.

Decentralized Model

Decentralized modeling is an important feature of this study. Previous data-driven methods usually develop a universal model through a machine learning algorithm, followed by fine-tuning hyperparameters that apply in various locations, assuming soil conditions can be represented by the training data and inferred from the developed model. While such applications have the benefit of wide implementation using a single model, they may have the risk of over-fitting to certain locations and conditions due to the various representativeness of every location in the training data. Oftentimes, a location contributes more quantity, and better quality of data can be modeled better over other places and may override local performances, impairing the inference accuracy. On the other hand, underperformance may exist if a high volume of low-quality data in a place pollutes the training data. Additionally, once new data is obtained, retraining and recalibration must be conducted to replace the entire model, causing uncertainties in maintaining the consistent performance in places where data was not updated. In this study, we propose decentralized models that individually develop a model for every single location

and every soil layer using their specific data, so that the risk of overfitting can be minimized. This may also simplify the updates of the model since only the locations having new data need to be retrained and recalibrated, not affecting the entire system and reducing the risks of inconsistent system performance.

Limits and Solutions

The limitations for this study span from the conditions for assumption-making, application cases, and the decentralized model design. The algorithm made several assumptions in designing the calculation model, such as no horizontal water movement, only modeling initial and end points of an increase event, isolated calculation for each soil layer, increase event division, etc. These are based on the sandy soil type in the specific location where the study is focused. With a high infiltration capacity in sandy soil, the soil moisture increase process is usually fast and short, which makes the impacts from the environment of a limited magnitude that can be ignored and narrows the concentration of the calculation on soil moisture increase and percolation. As soil sensors are placed at certain depths, continuous simulation of soil moisture over the depth is not easy to realize, so that every single soil layer needs to be modeled individually with the percolation parameters specifying interlayer connections. Event-based modeling design divides time series soil moisture records into increasing and decreasing sequences since their different hydrological processes (e.g. percolation) may require different calculation methods. Separate modeling upon the direction of soil moisture change allows simple and accurate representations of underlying hydrological processes. Such simplification also resulted in soil moisture fluctuations within an increase event not being important, as the irrigation decision-making will not respond at that detailed level. However, if soil types change or the infiltration capacity of the soil decreases so that the consideration of vertical movement of water cannot well embody the actual soil moisture response, all these assumptions could be void. In addition, ET is neglected during the relatively short soil moisture wetting phase, this assumption is reasonable for Florida weather pattern, but when adopted this model into another states with different weather patterns, this assumption will be void.

Additionally, this model is designed for irrigation decision-making. Many simplifications applied in modeling rely on the specific needs of irrigation, such as modeling linear change of soil moisture, ignoring the soil moisture fluctuations in fine temporal scales, validating only the alignments of event duration and soil moisture change amount, and data from one weather station representing a relatively large surrounding area. This makes the model not suitable for detailed requirements of soil moisture, such as urban stormwater management.

The decentralized model mechanism also has its limitations. Having a single model for every location and every soil layer, the management of these models could be complex when there are many locations included with a higher

spatial resolution applied. An additional layer of identifying the proper model to be used is needed to facilitate various users' requests. While a model may represent the surrounding areas in similar soil and environment conditions, isolated locations with no previous data may not be well modeled since such a mechanism was not designed for high capability of inference capability. Yet, once applied over time and more locations are included in the system, the enhanced coverage of models may offset the limited inference capacity.

Impact on Future Development

For future development impact, this decentralized modeling approach enables future modeling approaches to use federated learning or edge computing to train local soil moisture models at their localized field with in-situ sensors and then update shared model parameters with a central server (Wen et al., 2023). This preserves privacy and allows the global model to benefit from diverse, location-specific data. For irrigation, this means each farm could adapt the predictive model to its soils and microclimate, while still learning from regional data resources (Ahmadi & Mahashabde, 2024). This decentralization design allows the future model to be more scalable and responsive to heterogeneity across different soil types in different landscapes.

In addition, this approach merges the physical rules into data-driven modeling. While "blackbox" data-driven models are easy to develop and can provide good performance, future agriculture may require higher explainability of data-driven models for trust and responsibility. By embedding known hydrologic principles, such as mass conservation and vertical water redistribution, into the XGBoost model ensemble through layered water balance equations, this study is not only learning from data correlations but is instead guided by tailored hydrological modeling to inject physically meaningful pathways. This constraint-driven learning path narrows the model solution space and enables it to find relationships that are consistent with soil physics rather than purely algorithmically finding patterns from data. As a result, the model not only becomes more interpretable but also provides relatively consistent performance across various soil and weather conditions, even when data is limited.

Combining these two features, future development of this system will focus on enhancing the coverage of the system by obtaining data of heterogeneous conditions of soil, weather, ecosystem, cropping operations, etc. Integrated with the component study of soil drying pattern profiling (Zhang & Yu, in progress), a service can be provided to customers to build their localized models to better inform their irrigation practices.

Conclusion

This study addressed the challenges in modeling RZSM, particularly the soil moisture increase phase as water is input into the soil. By developing an interpretable, event-based modeling framework that integrates layered water

balance equations with an ensemble of XGBoost models, this study provides a robust and physically consistent approach that bridges the gap between purely data-driven and physically based mechanistic models. The proposed model simulates water redistribution across multiple soil layers, capturing nonlinear soil dynamics during different water input intensities (heavy, medium, and light).

Results demonstrate the XGBoost model ensemble's ability to simulate the soil interlayer connection, particularly under heavy and medium event conditions. However, accuracy slightly decreased for light events, especially in deep soil layers, reflecting the inherent complexities of modeling deep-layer soil moisture dynamics. The model ensemble has a gradient decrease in performance as soil depth increases, which is consistent with known hydrological principles that moisture infiltration signals become increasingly variable and challenging to predict deeper soil layers. The error distribution further analysis of this depth-dependent performance indicates better performance at soil moisture increase duration predictions at the surface, but more stable moisture increased amount predictions at the deep soil layer, which matches physical soil moisture behaviors as validated by recent literature.

This paper provides a practical, interpretable, and physically informed solution to model the RZSM increase phase. The results underscore the effectiveness of using an XGBoost model ensemble within a layered water balance framework, using hydrological rules to guide the data-driven model learning path. This approach is suitable for agriculture applications, particularly in irrigation scheduling, for future integration of event-based machine learning models in soil moisture prediction across different agricultural contexts.

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Ethical Considerations

This article does not contain any studies with human or animal participants.

Consent to Participate

There are no human participants in this article and informed consent is not required.

Author Contributions

All authors significantly contributed to the study's design, data collection, or analysis; drafted or critically revised the article; and

approved the final version. All authors agree to be responsible for all aspects of the work to address any issues regarding its accuracy or integrity. All contributors who do not meet the criteria for authorship has been included in the Acknowledgments section.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix

Sensors and Soil Moisture Data Collection

This project uses BMP Logic soil moisture probes (Sentek Drill & Drop™, Sentek Technologies, Adelaide, Australia) for soil moisture data collection. Drill & Drop probes are multi-depth capacitance sensors with multiple sensing elements to detect the change in electrical capacitance and converted to volumetric water content using a calibration equation. The probes installed underground were 24 in (60 cm) with sensing elements spaced at 4 in (≈10 cm) intervals. Because each sensing element integrates soil permittivity over a finite volume surrounding its nominal depth, we interpret these three depths as representative of the 0 to 2, 2 to 6, and 6 to 10 in layers when converting volumetric water content to soil water amount (mm) for 2, 6, and 10 in soil layer. In Florida’s sandy vegetable systems, most sweet corn roots responsible for water uptake are concentrated within the upper 25 to 30 cm (≈10–12 in). Therefore, this study using 2, 6, and 10 in as three effective soil layers for modelling.

In this study, “soil moisture” refers to the volumetric water content ($\text{m}^3 \text{m}^{-3}$) measured by the sensors. We convert volumetric soil moisture to “soil water amount,” defined as the equivalent water depth (mm) obtained by multiplying the volumetric content by the thickness of the corresponding layer (e.g. 0–2, 2–6, and 6–10 in). Unless otherwise specified, all results are reported in terms of soil water amount (mm).

Precipitation Event Determination

The precipitation events are determined by an inter-event dry period. There is a paper (Tu et al., 2023) stated that grouping successive water inputs within a 3-hr gap and a minimum of 0.5 mm amount is supported by hydrological event separation theory. Similarly, a study (Ignaccolo & Michele, 2009) mention a 1.5 to 4 hr no rain time range can define distinct rainfall events. Within this range, a 3 hr gap corresponds to the approximate time needed for soil infiltration processes between precipitation events and there are many hydrological studies (Pérez Bello et al., 2022; Tamm et al., 2023; Visser et al., 2021) adopt the 3 hr as a standard minimum inter-event time. Therefore, the soil moisture increase event in this study is defined as a new water input occurring at least 3 hr after the last water input event and with a minimum threshold of 0.1 mm of water input to avoid potential sensor noise. Although this 0.1 mm water input amount threshold is lower than the commonly used 0.5 mm, it is chosen because rain gauges from weather stations typically have a resolution of 0.1 to 0.2 mm, and water inputs below 0.1 mm are often considered unreliable due to

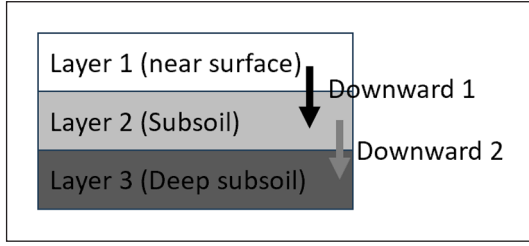


Figure 9. Soil moisture percolations through three soil layers. Note. Water infiltrates from Layer 1 (near surface) into Layer 2 (subsoil), represented by “Downward 1,” and subsequently from Layer 2 into Layer 3 (deep subsoil), indicated as “Downward 2.” Each layer represents a progressively deeper soil zone involved in water movement.

instrumentation limitations (Habib et al., 2001; Quintana-Seguí et al., 2008). This definition ensures that temporally clustered water inputs within a 3-h window are treated as a single water input event, while inputs separated by longer intervals are detected as distinct water input event.

Water Balance Equation for Layer 2 (Subsoil, 6 in)

Figure 9 shows the hydrological processes that flow through layer 2 (subsoil), as water moves from layer 1 (near surface) to Layer 3 (deep subsoil). Water percolates from the near surface layer 1 enter layer 2 is Downward 1, increasing soil moisture content in the intermediate subsoil. Depending on the antecedent moisture conditions, Layer 2 may retain all or part of the downward water from layer 1. If the retained water exceeds the field capacity of layer 2, the excess will percolate further into layer 3 as Downward 2.

The water balance equation for layer 2 is more complex than layer 1, even though the big picture still focuses on the water amount conservation. Layer 2 water balance equation is shown in equation (6). The left side of equation (6) is water transferred from layer 1, which represents the total amount of water into layer 2. The right side of equation (6) is water transferred to deeper layer 3 (downward 2), and water remains in layer 2 ($\theta_{2,end} - \theta_{2,init}$).

$$Downward_1 = Downward_2 + (\theta_{2,end} - \theta_{2,init}) \quad (6)$$

In practice, $Downward_1$ is known information can be calculated based on equation (5). The $\theta_{2,init}$ is provided by the dry phase model (layer 2).

The input parameters to predict the $\theta_{2,end}$ is like the equation (2),

$$x_2 = \{\theta_{2,init}, Downward_1, \Delta \hat{t}_1\} \quad (7)$$

Where $\theta_{2,init}$ is layer 2 antecedent soil moisture condition, $Downward_1$ is the water percolates into layer 2.

Based on the input parameters x_1 , investigate a model $f(x_2)$ to predict $\hat{y}_2$: the layer 2's soil moisture condition after percolation diminished.

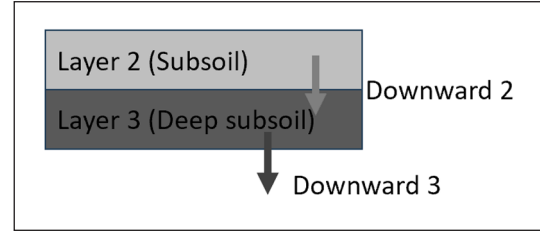


Figure 10. Continued percolation from subsoil to deep subsoil layers.

Note. Moisture moves downward from Layer 2 (subsoil) to Layer 3 (deep subsoil), marked as “Downward 2,” and further percolates beyond the deep subsoil as indicated by “Downward 3.”

The model predicts two outputs:

$$\hat{y}_2 = f(x_2), \text{ where } \hat{y}_2 = \begin{bmatrix} \Delta \hat{\theta}_2 \\ \Delta \hat{t}_2 \end{bmatrix} \quad (8)$$

Where $\Delta \hat{\theta}_2$ is the layer 2 soil moisture increased amount, $\Delta \hat{t}_2$ is the layer 2 soil moisture increased duration, which represents the percolation time.

Then the $\theta_{2,end}$ can be calculated as:

$$\theta_{2,end} = \theta_{2,init} + \Delta \hat{\theta}_2 \quad (9)$$

Submitted equation (5) and (9) into equation (6) gets:

$$Downward_2 = Water\ input - \Delta \hat{\theta}_1 - \Delta \hat{\theta}_2 \quad (10)$$

Intuitively, equation (10) represents the external water input minus the water remaining in layer 1 and layer 2 equals the downward outflow from layer 2 to layer 3.

The Water Balance Equation for Layer 3: Deep Subsoil (10 in)

Figure 10 shows the water flow dynamics between the subsoil (Layer 2) and deep subsoil (Layer 3). Deep subsoil layer 3 is the lower boundary of the active root zone considered in this study. Part of the water remained in layer 3, while other than the remaining water that exceeds the layer's retention capacity as downward 3.

The water balance equation for layer 3 is also based on the water conservation principle, but unlike the subsoil, layer 3 (10 in) is the deepest soil layer in this study. The layer 3 water balance equation accounts for vertical water transfer from layer 2 and flows out to the deeper layer but $Downward_3$ will not be used as more deep layer water input parameter.

Layer 3 water balance equation is represented as equation (11).

$$Downward_2 = Downward_3 + (\theta_{3,end} - \theta_{3,init}) \quad (11)$$

In practice, $Downward_2$ and $\theta_{3,init}$ is known parameters. From the modeling perspective for layer 3:

$$x_3 = \{\theta_{3,init}, Downward_2, \Delta \hat{t}_2\} \quad (12)$$

The model predicts two outputs:

$$\hat{y}_3 = f(x_3), \text{ where } \hat{y}_3 = \begin{bmatrix} \Delta\hat{\theta}_3 \\ \Delta\hat{t}_3 \end{bmatrix} \quad (13)$$

Where $\Delta\hat{\theta}_3$ is the layer 3 soil moisture increased amount, $\Delta\hat{t}_3$ is the layer 3 soil moisture increased duration, which represents the water percolation from layer 2 to layer 3 time.

Then the $\theta_{3,end}$ can be calculated as:

$$\theta_{3,end} = \theta_{3,init} + \Delta\hat{\theta}_3 \quad (14)$$

Review of Previous Modelling Methods for Soil Moisture

Southeast U.S., analyzing each location's soil dynamics individually is needed. Due to the specific localized requirement, the data availability is narrowed down from the state scale to the location scale, which brings limited data volume constraints for model training. Second, this study aims to capture soil moisture increase patterns across multiple layers under different soil and weather conditions, for example, what surface soil moisture condition and water input amount allows the water infiltration and redistribution into deeper soil at what rate and amount. This requires an approach to learning nonlinear relationships. Third, this study is intended for use by farmers and agricultural extension agents, so model interpretability is important. The model needs to be easy to understand, how it works, and how the in-field soil and weather data are involved in the model. These three constraints restrain many approaches that can be used.

Firstly, the intensive parameters needed to restrain the use of physics-based hydrological models. Like simulators that solve Richards' equation, either directly or via software packages such as HYDRUS or the Root Zone Water Quality Model (RZWQM), which can simulate soil-water interaction and root water uptake, but demand detailed site-specific information (soil hydraulic parameters, retention curves, root distributions) and careful calibration. Collecting and calibrating these inputs for each field in the study is impractical due to the cost and time of regular parameter assessments.

Secondly, the nonlinear relationship requirements restrain the use of traditional statistical models (e.g. linear regression models or moving average statistical approach). These models can work with limited data volume and are easy to interpret (Giorgio et al., 2022), but they assume linear or fixed input-output relationships, which do not allow them to capture nonlinear soil dynamics. However, some nonlinear regression models, such as polynomial or interaction regression, can capture nonlinear dynamics, but they require pre-specific interaction terms (e.g. $x_1 \times x_2$) or high-order terms (e.g. x^2 , x^3) that need to be arbitrarily selected, lacking the confidence of representing key interactions.

Thirdly, interpretability problems and large data volume requirements restrain the use of deep learning neural networks. Deep learning models are "black boxes," which are difficult to interpret the underlying structure and how the

hydrological process is represented. In addition, the deep learning approach needs large data volumes for training (Zabërgja et al., 2025), which cannot be met by this study.

In short, a deep learning approach is impractical with limited data volume and lack of interpretability. Linear regression is simple but cannot capture the local uncertainties in nonlinear relationships, and physics-based hydrological models require intensive parameters that are impractical for every individual site in this study. Instead, tree-structured machine learning methods, the Gradient-Boosted Decision Trees (XGBoost, LightGBM, CatBoost) are employed to overcome these constraints.

Recent studies show that Extreme Gradient Boosting (XGBoost) performs well on small to medium dataset volumes and particularly in environmental and engineering-related applications. For instance, (Chen & Guestrin, 2016; Lamane et al., 2025) states that XGBoost regularization and tree boosting structure make the model work and generalize well even as the dataset volume is limited.

In addition, XGBoost offers interpretable decision structures, making its use by farmers and extension agents practical. For example, (Islam et al., 2024) introduced iXGB, a method that extracts human-readable rules (e.g. "if soil moisture < X and rainfall > Y then deep-layer increase = Z") directly from trained XGBoost models, which explain what would happen if key inputs changed. Additionally, (Lundberg et al., 2019) developed an algorithm that integrates into XGBoost and allows users to see how much each variable influenced a given prediction.

Finally, XGBoost has been proven effective in extracting nonlinear soil moisture dynamics in the agriculture application. For instance, there are some studies have found that XGBoost has better performance compared with neural network, random forest, and SVM in predicting soil moisture at the state scale (Ren et al., 2023; L. Wang & Gao, 2025). Another study (Tahmouresi et al., 2024) using multi-sensor remote sensing combined with XGBoost gets the best performance among ensemble methods for retrieving soil moisture, which demonstrates that XGBoost is good at capturing complex moisture and environment relationships.

Hyperparameter Tuning

To obtain better performance of the model, the Bayesian optimization approach (Optuna) combined with the cross-validation method has been adopted to fine-tune the key hyperparameters of XGBoost in this study. Bayesian optimization is an efficient algorithm that finds cross-validated error (RMSE) for different hyperparameter combinations. Notably, this tuning process was applied to three XGBoost models for three soil moisture layers, which allows each layer's model to get its own optimal tuned hyperparameters.

Key hyperparameters and their roles in the model include:

- Number of Trees (n_estimators): The maximum number of boosting rounds (trees) to fit. Default value = 100.

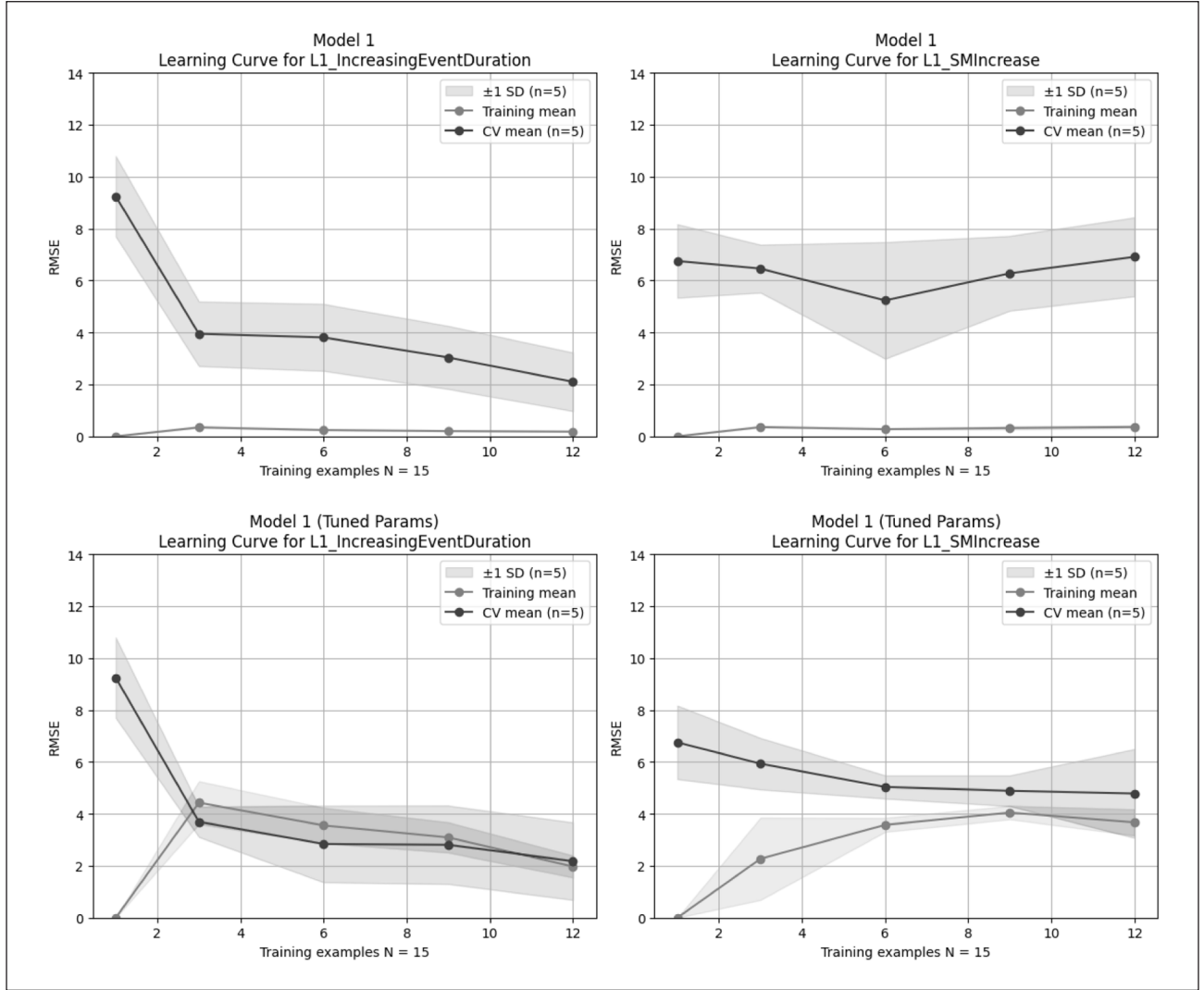


Figure 11. Learning curves of model performance for Layer 1 (2 in depth) before (top row) and after (bottom row) hyperparameter tuning using Bayesian optimization.

Note. The left panels show model predictions for the event duration (L1_IncreasingEventDuration), while the right panels show predictions for soil moisture increase amounts (L1_SMIIncrease). Each curve compares training scores (grey line) and cross-validation scores (black line), shaded bands indicate mean ± 1 standard deviation (SD) of the cross-validation score across folds ($n=k$, where k is the number of CV folds, $k=5$ in this study) and the total sample $N=15$ used for hyperparameter tuning for one location's model. Improved alignment and reduced gaps between training and cross-validation scores in the tuned model indicate enhanced model generalization and reduced overfitting.

- Learning Rate (η): A shrinkage factor ($0 < \eta < 1$) that scales the contribution of each new tree. Default value=0.3.
- Max Tree Depth: The maximum depth of each decision tree. This controls the model's complexity. Default value=6.
- Min Child Weight: The minimum sum of instance weights (also known as hessian, or second-order gradient) required to split a child node from a parent node in a tree. A child node is the result of a split from a parent node in a decision tree. Default value=1.
- Gammas: The minimum loss reduction required to make a further partition on a leaf node. It acts as a regularization parameter that controls tree complexity. Default value=0.
- Subsampling Fractions: applied row subsampling (subsample) and column subsampling (colsample_bytree) rates for each tree. Subsampling helps reduce overfitting by ensuring no single tree sees all the data or all features. The default value for subsample=1, colsample=1.
- Regularization Parameters: XGBoost provides L1 and L2 regularization on tree weights. Default value for L1=0, L2=1.

Result 1: Hyperparameter Tuning

Figure 11 illustrates the performance improvements achieved before and after applying Bayesian hyperparameter tuning to Model 1 (Layer 1) using $\Delta = RMSE_{CV} - RMSE_{Train}$ as the overfitting gap. The total sample $N=15$ is a relatively

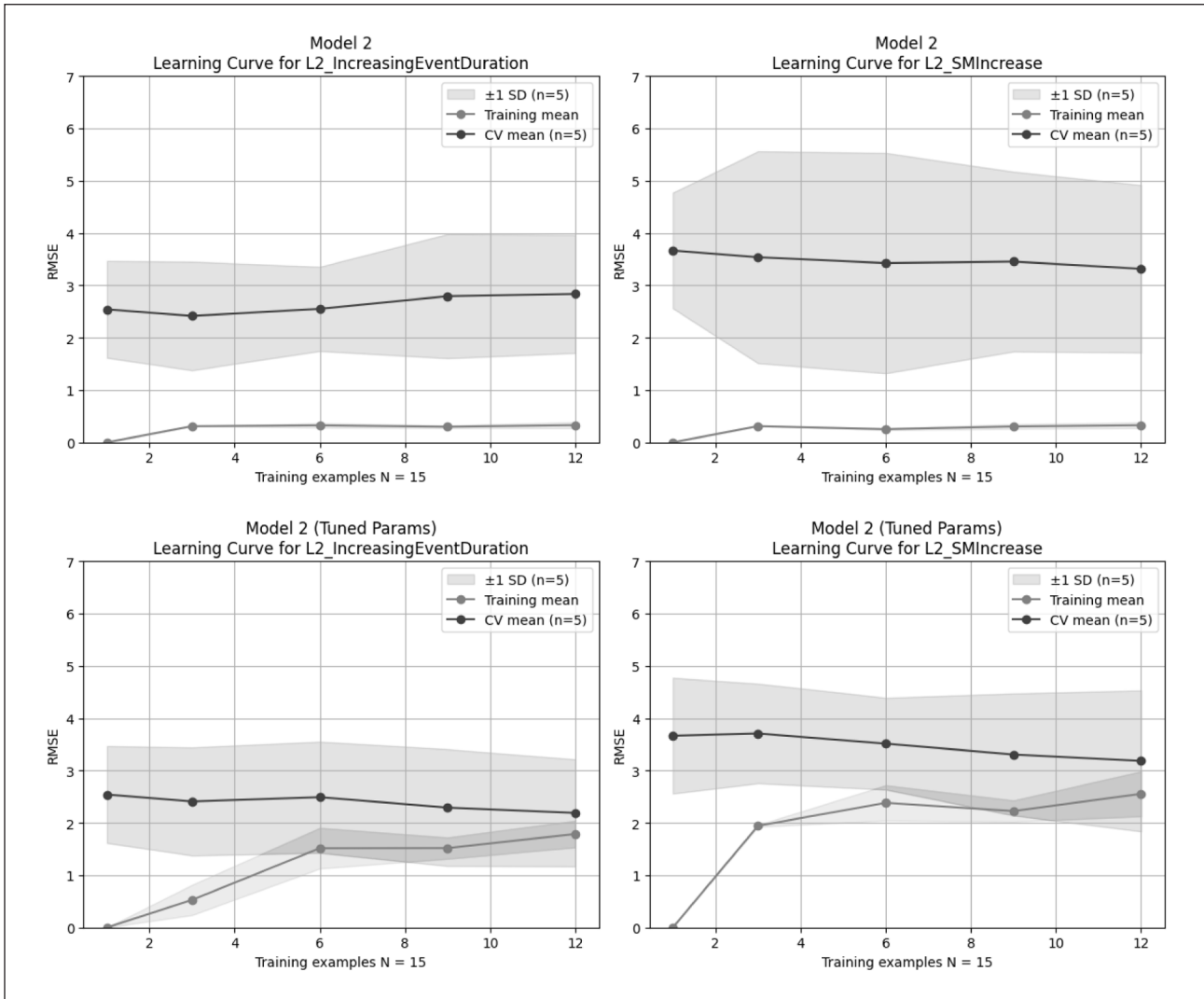


Figure 12. Learning curves showing model performance for Layer 2 (6 in depth) before (top row) and after (bottom row) hyperparameter tuning via Bayesian optimization.

Note. The left panels show predictions of event duration (L2_IncreasingEventDuration), while the right panels show predictions of soil moisture increase (L2_SMIncrease). Training scores (grey line) are compared to cross-validation scores (black line), shaded bands indicate mean ± 1 standard deviation (SD) of the cross-validation score across folds ($n=k$, where k is the number of CV folds, $k=5$ in this study) and the total sample $N=15$ used for hyperparameter tuning for one location's model. The tuned model shows reduced difference between training and cross-validation scores, especially the event duration, indicating improved generalization and minimized overfitting.

small amount for tuning parameter very well due to restricted by the 2year event data amounts and decentralized design further split the data volume to each location, so this section for hyperparameter tuning just for demonstration. For L1_SMIncrease, Δ drops as training example increased from about 1.8 (pre-tuning: CV ≈ 2.2 vs. train ≈ 0.4) to about 0.3 (tuned: CV ≈ 3 vs. train ≈ 3.3), around 80% reduction. For L1_IncreasingEventDuration, Δ shrinks from near 4.8 (pre-tuned: CV ≈ 5 vs. train ≈ 0.2) to near 0.7 (tuned: CV ≈ 4.5 vs. train ≈ 3.8), around 85% reduction. Those reductions mean the cross-validation curves are closer to the training curves after tuning, indicating reduced variance and better model generalization.

Figure 12 illustrates the impact of hyperparameter tuning on Model 2 in one of location's model for demonstration. For L2_IncreasingEventDuration, the gap at the

largest training size shrinks from ≈ 2.5 (pre-tuning: CV ≈ 2.9 vs. train ≈ 0.4) to ≈ 0.3 (tuned: CV ≈ 2.1 vs. train ≈ 1.8), around $\approx 85\%$ reduction. For L2_SMIncrease, the gap drops from ≈ 3.0 (pre-tuning: CV ≈ 3.3 vs. train ≈ 0.3) to ≈ 0.4 (tuned: CV ≈ 3.1 vs. train ≈ 2.6 – 2.7), around $\approx 82\%$ reduction. Cross validation score is stable or slightly lower after tuning, which indicates better regularization, reduced variance, and better generalization.

Figure 13 displays the effects of hyperparameter tuning on Model 3 in one of location's model for demonstration. After tuning, overfitting is diminished for Model 3 (10 in layer). The gap between training and validation $\Delta = RMSE_{CV} - RMSE_{Train}$ decreased as the training example increased for both two target parameters. For L3_IncreasingEventDuration, Δ drops from ≈ 2.3 (pre-tuning: CV ≈ 2.6 vs. train ≈ 0.3) to ≈ 0.3 (tuned: CV ≈ 2.4 vs. train ≈ 2.1), around 87% reduction. For

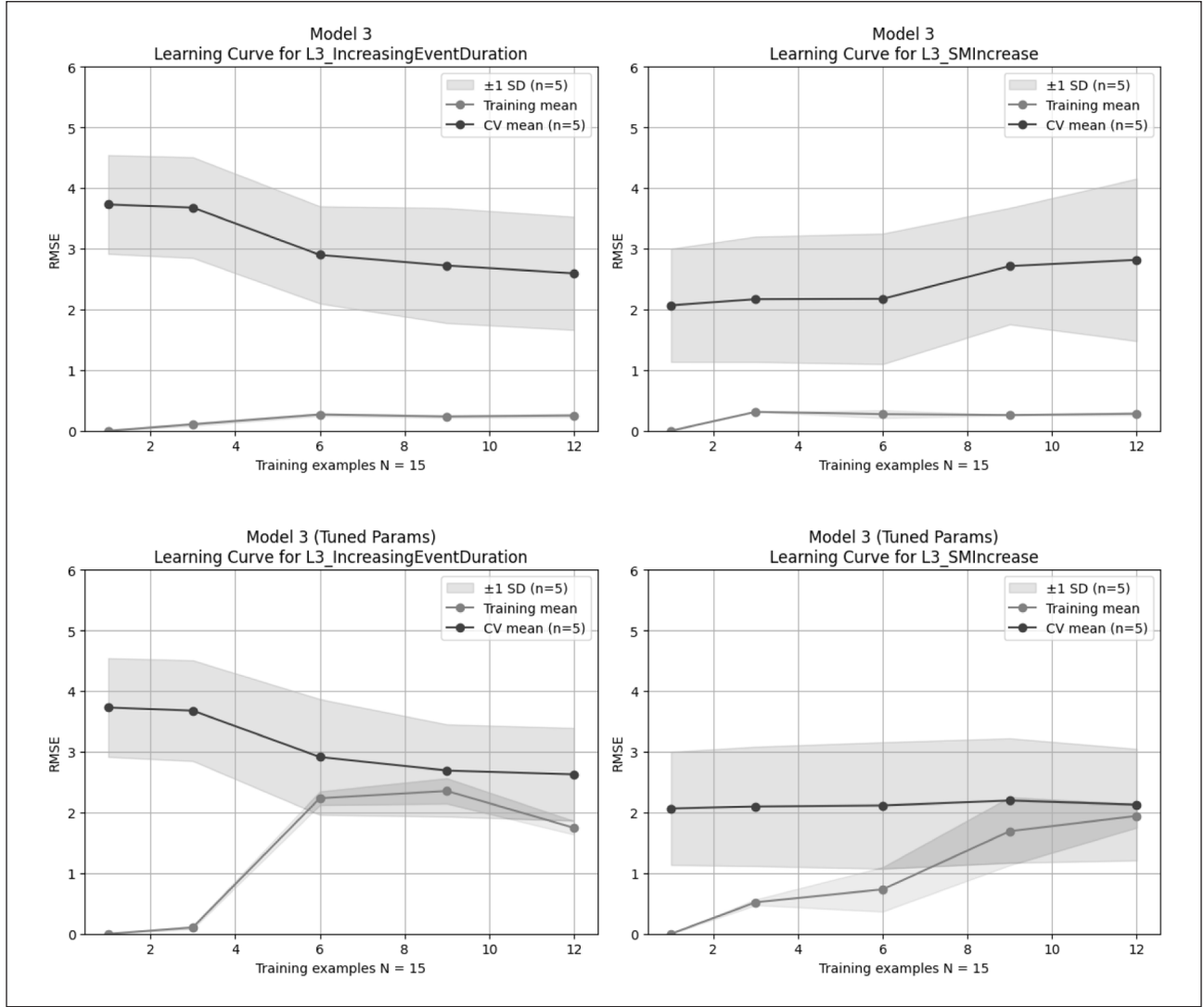


Figure 13. Learning curves of model performance for Layer 3 (10 in depth) before (top row) and after (bottom row) Bayesian hyperparameter tuning.

Note. Left panels represent predictions of event duration (L3_IncreasingEventDuration), and right panels represent predictions of soil moisture increase (L3_SMIncrease). Training scores (grey line) and cross-validation scores (black line). Each curve compares training scores (grey line) and cross-validation scores (black line), shaded bands indicate mean ± 1 standard deviation (SD) of the cross-validation score across folds ($n=k$, where k is the number of CV folds, $k=5$ in this study) and the total sample $N=15$ used for hyperparameter tuning for one location's model. After hyperparameter tuning, the curves exhibit improved convergence between training and cross-validation scores, highlighting better generalization and reduced overfitting for the deepest soil layer.

L3_SMIncrease, Δ falls from ≈ 2.5 (pre-tuning: CV ≈ 2.7 vs. train ≈ 0.2) to ≈ 0.2 (tuned: CV $\approx 2.0-2.1$ vs. train $\approx 1.9-2.0$), around 90% reduction. In summary, these changes indicate reduced model variance and improved generalization for the deepest soil layer model.

Hyperparameter Recommendation

To support future applications of the increased soil moisture prediction, this section provides a generalized combination of hyperparameters for future users. In total, there are 93 models (3 layers per location) established, with data from 31 field locations across Florida. Each model is individually tuned using Bayesian optimization to find the optimal XGBoost hyperparameters. The optimized

hyperparameters for each model are aggregated by layer to examine their frequency distributions across locations. The purpose of this aggregation is to identify commonly used hyperparameter values that show relative general performance in their location and use those common hyperparameters as recommendations for future users when adopt this method to a new farm field. Such a combination provides practical guidance for future users, particularly those with limited computational resources or short of experience in machine learning.

Figure 14 shows the distribution of the tuned $n_estimators$ values across 31 locations for three soil layers (Model 1 to Model 3). The $n_estimators$ control the XGBoost model training rounds. A high value makes the model complex and has the potential to overfit, while a lower value may cause

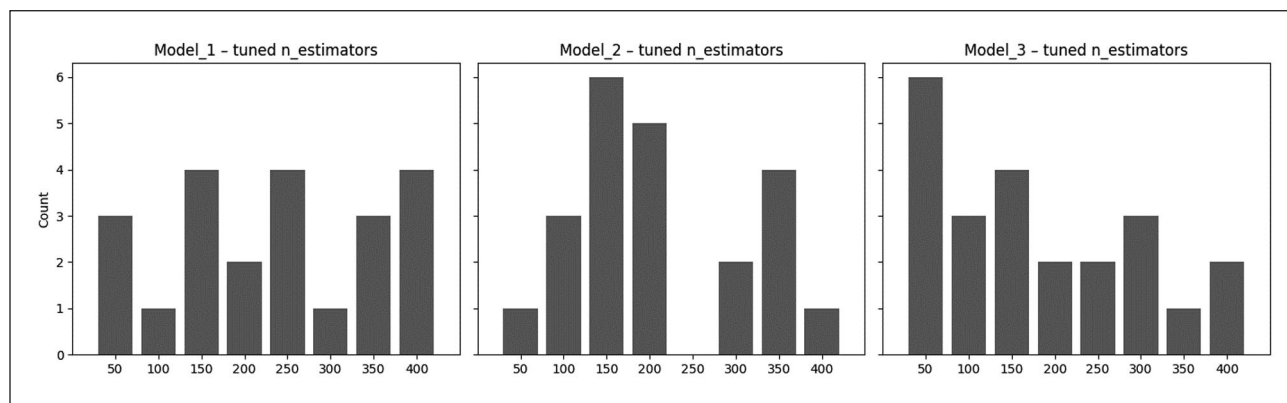


Figure 14. Distribution of optimized hyperparameter values for `n_estimators` across study locations for each soil layer model (Model 1:2 in, Model 2:6 in, and Model 3:10 in depth).

Note. Histograms illustrate the frequency of selected `n_estimators` values after Bayesian hyperparameter tuning. Recommended values for practical future applications were determined based on the most frequently selected values, providing balanced trade-offs between model complexity and predictive accuracy across different layers.

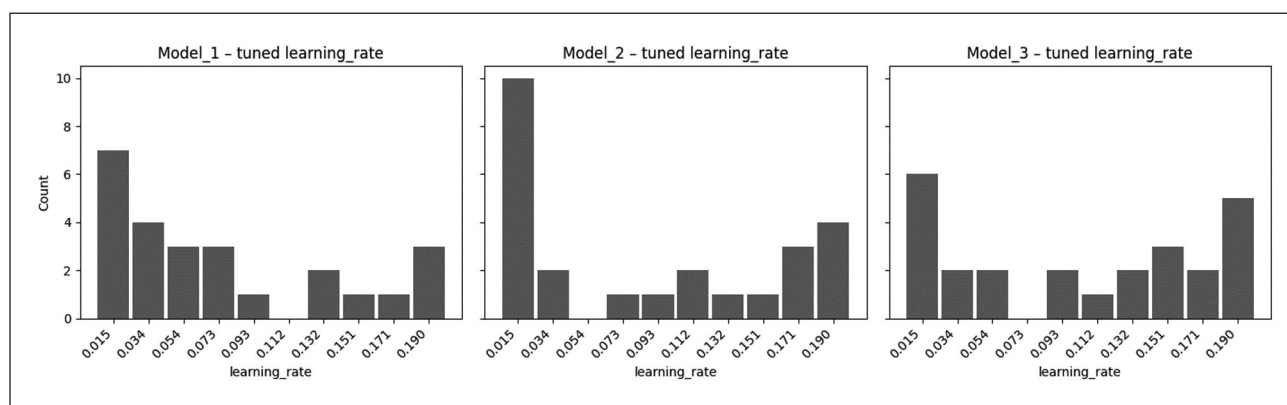


Figure 15. Distribution of optimized hyperparameter values for `learning_rate` across locations for each soil layer model (Model 1:2 in, Model 2:6 in, and Model 3:10 in depth).

Note. Histograms illustrate the frequency of selected `learning_rate` values after Bayesian hyperparameter tuning. A value of 0.015 consistently emerges as the most commonly selected rate, suggesting a preference for gradual learning processes that enhance generalization across various moisture conditions.

underfitting and reduce predictive performance. For Model 1 (Layer 1), `n_estimators` are distributed between 50 and 400, and no unique peak value, but depends on the same count for `n_estimators`=150, 250 and 400, we use the median value 250 as recommended value due to it has same counts and not too high to cause too much computation resources. Model 2 (Layer 2) shows the most common value, 150, with 6 counts, which suggests that intermediate boosting rounds provided the best trade-off between bias and variance. Model 3 (Layer 3) showed a concentration at lower values, 50 with 6 counts, which indicates that the lower training rounds are enough for the deep layer model. Based on the above analysis, the `n_estimators`' recommendations are: 250 for Layer 1, 150 for Layer 2, and 50 for Layer 3.

Figure 15 shows the distribution of the tuned learning rate. The learning rate controls the step size at each iteration. A smaller `learning_rate` will cause a slower but more robust learning process. As shown in the histograms, 0.015 is the most common value for the three models in three layers, which suggests that a small learning rate is generally suitable for modeling soil moisture dynamics in Florida's

diverse soil conditions. Therefore, `learning_rate`=0.015 is recommended.

Figure 16 shows the distribution of the tuned `max_depth`. The `max_depth` determines the maximum depth of each decision tree, which controls how complex the model can become when fitting the data. A larger `max_depth` allows the model to capture more complex data interactions, but with the risk increasement for overfitting. Model 1 and 2 shows that 3 and 9 are the most common values for `max_depth`, and the value 8 is most common in model 3. For Model 1 and Model 2 we selected `max_depth`=3, because the data volume for each location is limited, using a lower value as the `max_depth`=3 and `max_depth`=9 has same counts may help to diminish the potential overfitting. Therefore, the recommended value is: 3 for model 1 and model 2, 8 for model 3.

Figure 17 shows the distribution of tuned `min_child_weight`. The `min_child_weight` controls the minimum sum of instance weights (or data points) needed in a child node to split. Higher `min_child_weight` means the model needs more data to split a node, which thereby helps to avoid overfitting,

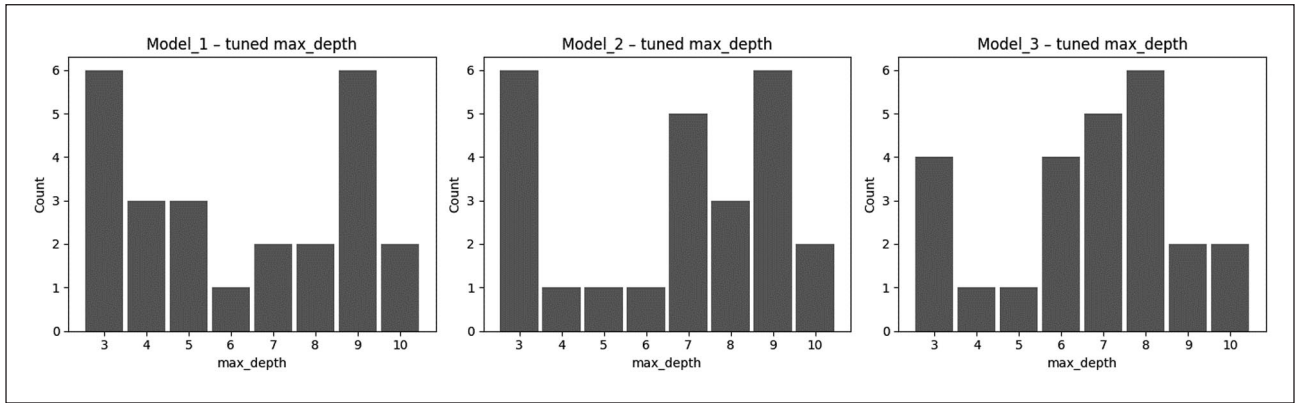


Figure 16. Distribution of optimized hyperparameter values for max_depth across locations for each soil layer model (Model 1:2in, Model 2:6in, and Model 3:10in depth).

Note. The histograms illustrate the frequency of max_depth values selected by Bayesian hyperparameter tuning. Optimal max_depth values tend to differ by soil layer, reflecting varying complexity in modeling moisture dynamics: shallower depths (Models 1 and 2) often favor simpler models (depth=3), while deeper layers (Model 3) prefer slightly greater complexity (depth=8).

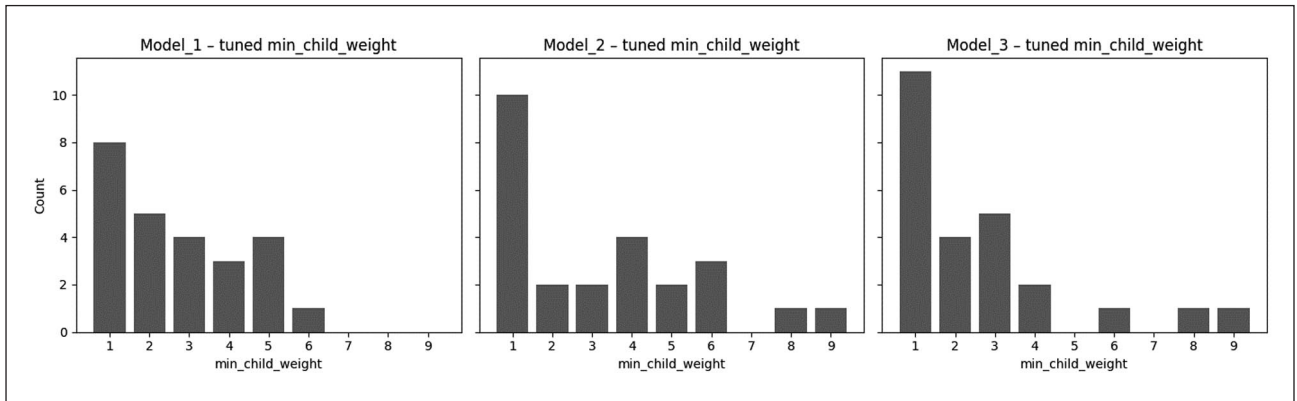


Figure 17. Distribution of optimized hyperparameter values for min_child_weight across locations for each soil layer model (Model 1:2in, Model 2:6in, and Model 3:10in depth).

Note. Histograms display the frequency of selected min_child_weight values from Bayesian optimization. A clear preference for a minimal weight value of 1 emerges consistently across all soil layers, suggesting that simpler splits with fewer data points per leaf node effectively balance model complexity and generalization in soil moisture predictions.

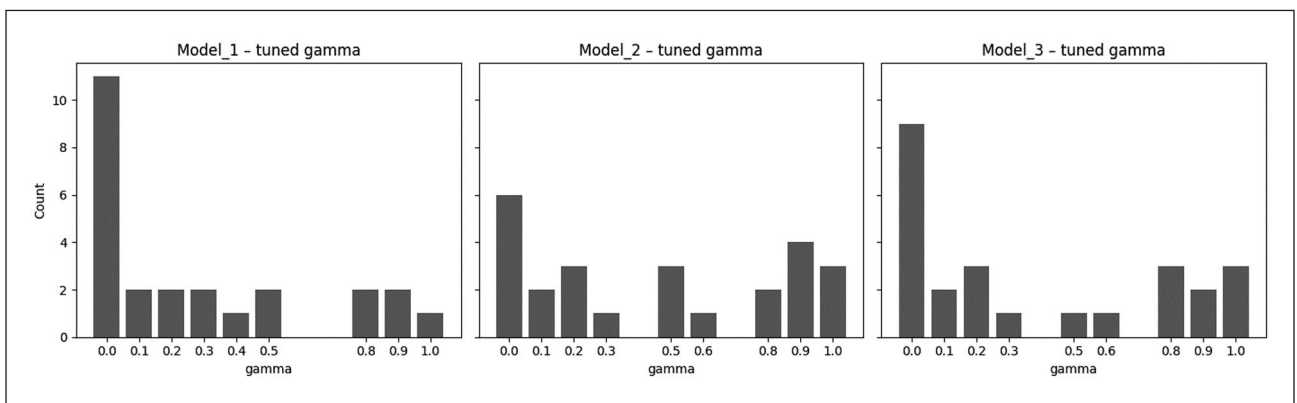


Figure 18. Distribution of optimized values for the gamma hyperparameter across locations for Models 1, 2, and 3 (corresponding to 2, 6, and 10in soil layers, respectively).

Note. Gamma controls the minimum loss reduction required for tree splitting. A value of 0 was most frequently selected across all models, indicating that minimal penalization for splits yielded better performance under varying soil moisture conditions.

but requires the dataset in large volume. From the histograms, min_child_weight=1 is the most common value for all models. Therefore, the recommended default setting for min_child_weight would be 1 for all three layers.

Figure 18 shows the distribution of the tuned gamma. Gamma represents the minimum loss reduction required to make a new partition on a tree's leaf node, which controls the model's split sensitivity. As shown in the histograms,

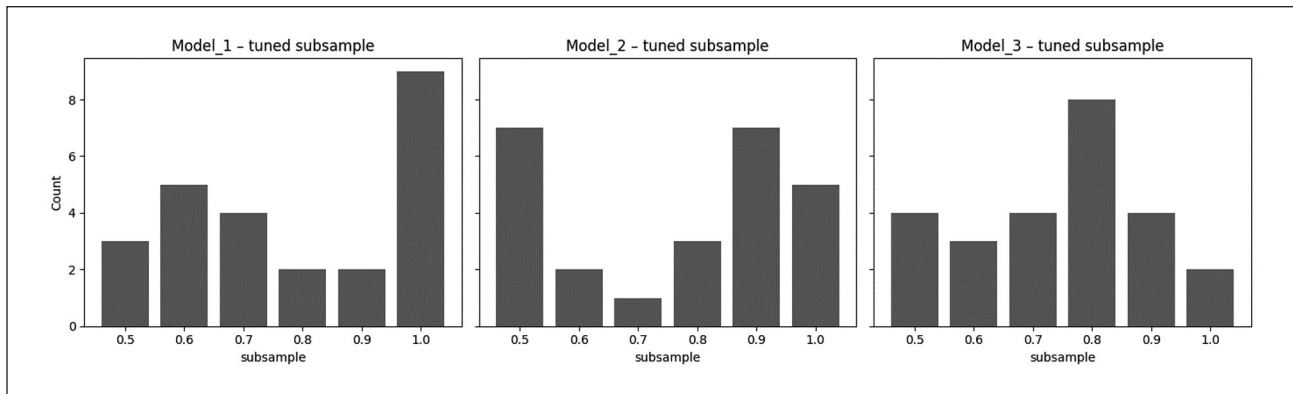


Figure 19. Distribution of tuned subsample values across locations for each soil layer model. Subsample defines the fraction of training samples randomly used to grow each tree.
Note. Model 1 (surface layer) favored higher subsample values (especially 1.0), while Models 2 and 3 (subsurface layers) showed greater variability, suggesting that medium sampling improved generalization in deeper layers.

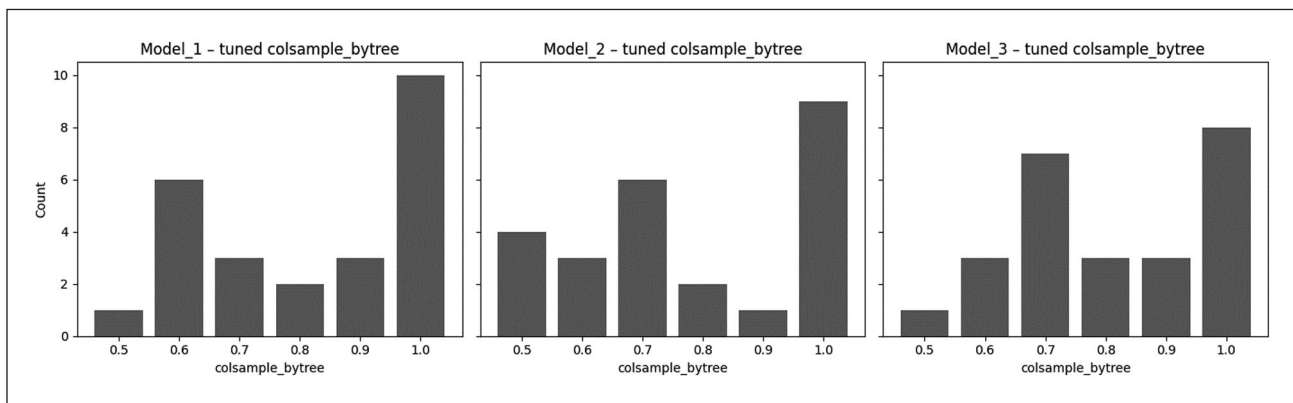


Figure 20. Distribution of tuned colsample_bytree values across locations for each soil layer model.
Note. This hyperparameter determines the fraction of input features randomly selected for each tree. A strong preference for using value = 1.0 was seen across models, supporting the use of full feature sets for reliable soil moisture prediction.

gamma=0 is the most common value in all three models, which indicates that no penalty splits provide better performance in most locations. Therefore, gamma=0 is recommended for all three models.

Figures 19 and 20 show the distributions of the tuned subsample and colsample_bytree, respectively. The subsample defines the fraction of training data (rows) randomly selected for each tree, and the colsample_bytree controls the fraction of features (columns) randomly selected for each tree. These two hyperparameters play complementary roles in controlling the amount of data and features used in the model training. For Figure 19, 1.0, 0.9, and 0.8 are the most common subsamples for models 1, 2, and 3. For Figure 20, 1.0 is the most common subsample in all three models, which indicates that using nearly all

features may lead to optimal performance. Therefore, the recommendation for subsamples is 1.0 for model 1, 0.9 for model 2, 0.8 for model 3, and the recommendation for colsample_bytree is 1.0 for all models.

As shown in Figures 21 and 22, the distributions of reg_alpha and reg_lambda indicate that the L1 and L2 regularization are consistently tuned to very small values in all models, which means that regularization only provides little or no effect during training to diminish the overfitting.

The final hyperparameter recommendation values presented in Appendix Table 4 were derived from the tuning results as shown in Figures 14 through 22. Each figure visualizes the distribution of a specific parameter on model performance across different soil layers and locations.

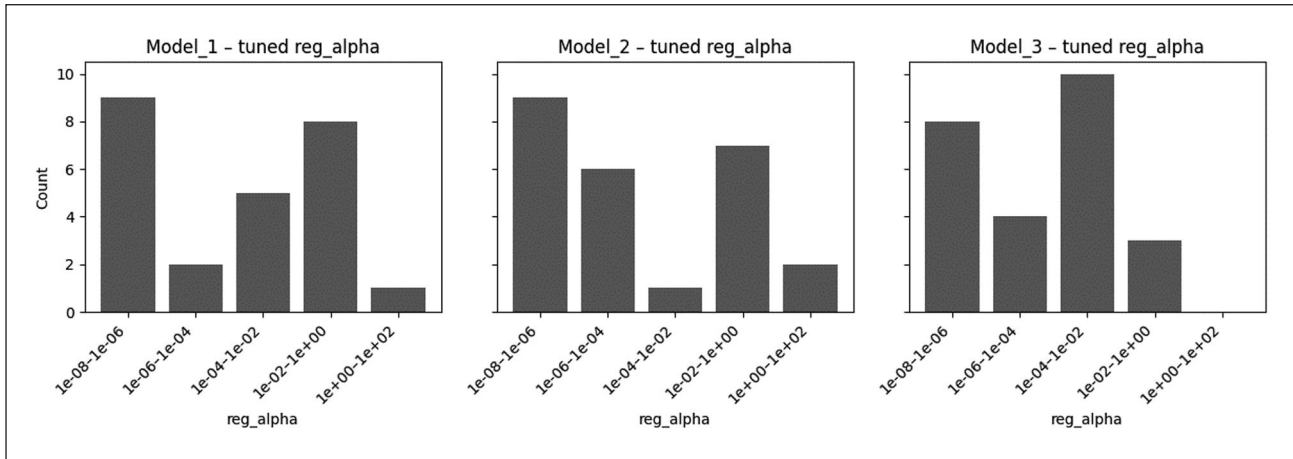


Figure 21. Distribution of tuned reg_alpha (L1 regularization) values for Models 1, 2, and 3.

Note. Most optimal values clustered around very small magnitudes (1e-03 to 1e+00), suggesting that L1 regularization had minimal impact in penalizing tree complexity and that sparse feature selection was not a dominant factor in model performance.

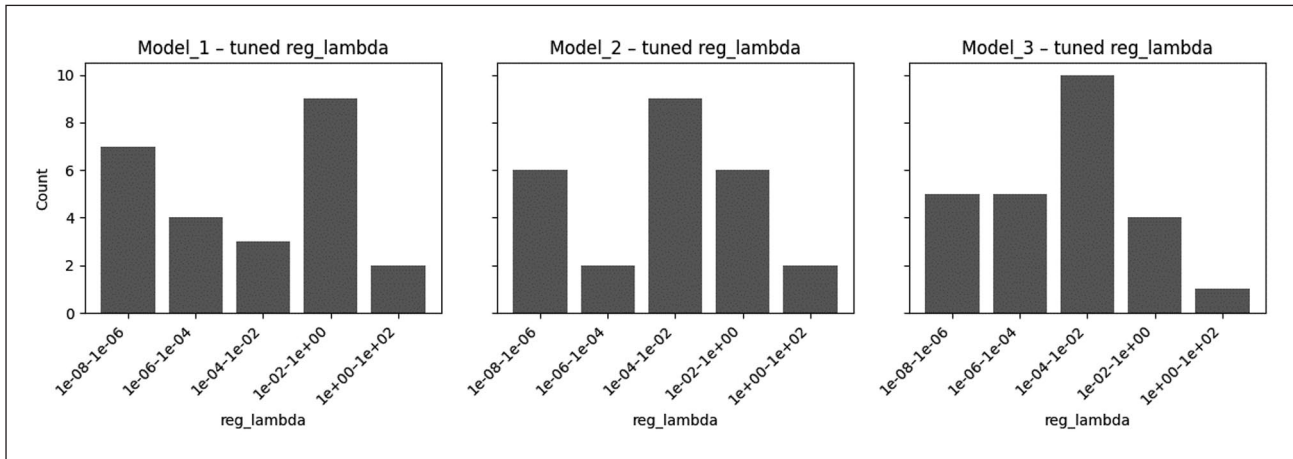


Figure 22. Distribution of tuned reg_lambda (L2 regularization) values across all three soil layer models.

Note. While optimal values varied more than those for reg_alpha , most selected values were still within a low to medium range (1e-04 to 1e+00), indicating that light L2 regularization was necessary to reduce overfitting in most locations.

Table 4. Hyperparameter Recommendation.

	$n_estimators$	Learning_rate	Max_depth	Min_child_weight	Gamma	Subsample	Colsample_bytree
Model 1	250	0.015	3	1	0	1	1
Model 2	150	0.015	3	1	0	0.9	1
Model 3	50	0.015	8	1	0	0.8	1